

Master Thesis

Beyond Goldfish Memory: What is Relevant in Long-Term Dialogue?

Carmen Ioana Popa

Supervisor Piek Vossen
2nd reader Pia Sommerauer

a thesis submitted in fulfillment of the requirements for the degree of

MA Linguistics
(Language and AI)

Vrije Universiteit Amsterdam

Computational Linguistics and Text-Mining Lab
Department of Language and Communication
Faculty of Humanities

Date July 3, 2026
Student number 2901935
Word count 27248

Abstract

In this thesis, we investigate which linguistic cues are informative enough to indicate relevance in a multi-turn conversation. The study focuses on the MSC:PersonaSummary dataset created by Facebook AI. It is a human-human, multi-session dialogue in which the crowd workers were asked to interact without providing any personal information, using only the persona assigned to them across the different sessions, with each session separated by a simulated time gap. The aim was to mimic the kind of chat that takes place on social platforms on a daily basis. Drawing on linguistic theories such as relevance theory, common ground and grounding, and the given-new contract, we used feature engineering to translate these linguistic cues into features through several techniques, and trained and evaluated a linear Support Vector Machines (SVM) model Cortes and Vapnik (1995) to predict which utterances are relevant for long-term conversation and memory. We chose an SVM model in order to offer a transparent and interpretable view of how relevance is selected. A feature ablation study was conducted to determine the importance of each linguistic cue under consideration. It showed that agency phrases and possessive phrases are the most important indicators of relevance, while emotion polarity proved to add noise to the performance of the SVM model.

In addition to the experiments conducted with the SVM model, we performed a human validation, prompt engineering with different LLMs, and BERTScore evaluation to assess the quality of the model's output against the 'gold' represented by the extracted persona. In the human validation, we examined closely whether the 'gold' aligns with the utterances, although this was done on a small set of samples from the test dataset. Our strongest model, which combines the linguistic cues with the text utterance, reaches a macro-F1 of about 0.75 on the test dataset, well above the majority-class floor of roughly 0.40.

Overall, the thesis contributes a transparent and interpretable framework for studying the linguistic cues that signal which information in a human-human conversation is relevant and worth retaining in long-term interaction by LLMs for more human-like conversations.

The code used during the thesis can be found on: GitHub CLTL Students Repository.

Declaration of Authorship

I, Carmen Ioana Popa, declare that this thesis, titled *Beyond Goldfish Memory: What is Relevant in Long-Term Dialogue?* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: July 3, 2026



Signed:

Acknowledgments

I would like to express my sincere gratitude to my supervisor, Piek Vossen, for accepting me as a student and guiding me through the thesis. You were always available for help, and your constant feedback and encouragement helped me shape the ideas and clarified my confusions. I would also like to thank my second reader, Pia Sommerauer, for the feedback and for taking the time to review my work. I am thankful to Lucia Donatelli for guiding me at the start of my thesis.

Finally, I am grateful to my family for their support and being patient with me in this journey.

Contents

Abstract i

Declaration of Authorship iii

Acknowledgments v

List of Figures ix

List of Tables xiii

1 Introduction 1
1.1 Research Questions . . . 1

2 Related Work and Linguistic Theories 3
2.1 Related Work . . . 3
2.2 Linguistic and Pragmatic Perspectives . . . 5
2.3 Relevance and Common Ground . . . 6
2.3.1 Relevance Theory . . . 6
2.3.2 Common ground . . . 7
2.4 Discourse markers as linguistic cues . . . 8
2.4.1 Given-New Contract Theory . . . 9

3 Dataset Exploration and Analysis 11
3.1 Dataset Description . . . 11
3.2 Dataset Statistics and Analysis . . . 13
3.2.1 General Statistics . . . 13
3.2.2 Macro Statistics - Training Dataset . . . 15
3.2.3 Micro (Linguistics) Statistics - Training Dataset . . . 18
3.2.4 Proportions Across Sessions - Training Dataset . . . 25

4 Methodology and Experiments 27
4.1 Feature Description . . . 27
4.2 Hypothesis . . . 27
4.3 Modeling . . . 28
4.4 Utterance Level Evaluation - Validation Dataset . . . 31
4.5 Utterance Level Evaluation - Test Dataset . . . 33
4.6 Session Level Evaluation - Validation and Test Dataset . . . 35
4.6.1 LLM as a Judge . . . 35
4.6.2 BERTScore . . . 36

4.7 Feature Ablation - Validation Dataset . . . 37
4.8 Feature Ablation - Test Dataset . . . 41
4.9 Error Analysis . . . 44
4.9.1 Emotion Polarity Feature Validation . . . 46
4.9.2 Human Validation . . . 48
4.9.3 Annotation Errors . . . 49
4.10 Predicting Relevance per Utterance with LLM . . . 51

5 Discussion **53**
5.1 Limitations of the Study . . . 55

6 Conclusion **57**
6.1 Future Work . . . 58

A Supplementary Material **61**

B Supplementary Material **65**
Data Statement (Bender and Friedman (2018)): Multi-Session Chat Dataset (MSC) . . . 65
LLM Prompts for summary session evaluation . . . 67
Summary-evaluation prompts (validation set) . . . 67
Summary-evaluation prompts (test set) . . . 68
Prompt Emotion Classification . . . 68
Prompt Utterance Relevance Prediction . . . 71
Sampled Utterances . . . 71

References **85**

List of Figures

4.1 Overview of the methodology 28
4.2 The top five positive and negative cue-text n-gram features for the relevant class, with their SVM coefficients. 46

A.1 Classification report for the first experiment (bag-of-words baseline) on the validation dataset. 61
A.2 Classification report for the second experiment (structured cues) on the validation dataset. 61
A.3 Classification report for the third experiment (structured cues plus cue n-grams) on the validation dataset. 62
A.4 Classification report for the fourth experiment (combined full model) on the validation dataset. 62
A.5 Classification report for the first experiment (bag-of-words baseline) on the test dataset. 62
A.6 Classification report for the second experiment (structured cues) on the test dataset. 63
A.7 Classification report for the third experiment (structured cues plus cue n-grams) on the test dataset. 63
A.8 Classification report for the fourth experiment (combined full model) on the test dataset. 63

List of Tables

3.1 Utterance Dataset Columns 12
3.2 Session Dataset Columns 13
3.3 General session-level and utterance-level statistics across dataset splits. . 14
3.4 Distribution of turns per session across dataset splits. 14
3.5 Per-session utterance counts and summary extraction across dataset splits. 14
3.6 Distribution of sentence counts, per utterance (raw conversations) and per session summary. 16
3.7 Average number of turns by simulated time gap between sessions. 18
3.8 Distribution of the negation particles within the negated utterances. . . 22
3.9 Distribution of temporal states across the utterances. 23
3.10 Frequency of the extracted temporal-deixis tokens. 24
3.11 Per-session statistics (part 1 of 2). 'Total turns' is a count; 'mean word count' is an average; the remaining columns are percentages. 25
3.12 Per-session statistics (part 2 of 2). All values are percentages. 25

4.1 Overview of the linguistic cues and features and their theoretical justification. 27
4.2 MSC Utterances Dataset Example 29
4.3 Extracted Features Example 30
4.4 Utterance-level results overview on the validation set (macro-averaged). 32
4.5 Per-class utterance-level results on the validation set. 32
4.6 Utterance-level results overview on the test set (macro-averaged). 33
4.7 Per-class utterance-level results on the test set. 34
4.8 Session-level LLM-as-a-judge scores on the validation set. *Arrows compare each session with the overall mean (↑ above, ↓ below).* 36
4.9 Session-level LLM-as-a-judge scores on the test set. *Arrows compare each session with the overall mean (↑ above, ↓ below).* 36
4.10 Session-level BERTScore results on the validation set. *Arrows compare each session's F1 with the overall mean (↑ above, ↓ below).* 37
4.11 Session-level BERTScore results on the test set. *Arrows compare each session's F1 with the overall mean (↑ above, ↓ below).* 37
4.12 Feature ablation on the validation set: macro-averaged metrics by experiment. *Arrows: each experiment's macro F1-score vs. experiment 3 (↑ higher, ↓ lower, → similar).* 38
4.13 Single-feature ablation on the validation set: macro-averaged metrics. *Note: the validation dataset contains 25,459 utterances in total, of which 16,432 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each feature's macro F1-score alone vs. experiment 3 (↑ higher, ↓ lower, → similar).* 38

4.14 Single-feature ablation on the validation set: relevant-class metrics. *Note: the validation dataset contains 25,459 utterances in total, of which 16,432 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each feature's 'relevant'-class F1-score alone vs. experiment 3 (↑ higher, ↓ lower, → similar).* 39

4.15 Leave-one-feature-out ablation on the validation set: macro-averaged metrics. *Note: the validation dataset contains 25,459 utterances in total, of which 16,432 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: change in macro F1-score when each feature is removed, vs. experiment 3 (↑ improvement, ↓ decline, → negligible).* 40

4.16 Leave-one-feature-out ablation on the validation set: relevant-class metrics. *Note: the validation dataset contains 25,459 utterances in total, of which 16,432 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: change in the 'relevant'-class F1-score when each feature is removed, vs. experiment 3 (↑ improvement, ↓ decline, → negligible).* 40

4.17 Feature-subset ablation on the validation set: macro-averaged metrics. *The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each subset's macro F1-score vs. experiment 3 (↑ higher, ↓ lower, → similar).* 41

4.18 Feature-subset ablation on the validation set: relevant-class metrics. *Note: the validation dataset contains 25,459 utterances in total, of which 16,432 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each subset's 'relevant'-class F1-score vs. experiment 3 (↑ higher, ↓ lower, → similar).* 41

4.19 Single-feature ablation on the test set: macro-averaged metrics. *Note: the test dataset contains 24,375 utterances in total, of which 16,397 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each feature's macro F1-score alone vs. experiment 3 (↑ higher, ↓ lower, → similar).* 42

4.20 Single-feature ablation on the test set: relevant-class metrics. *Note: the test dataset contains 24,375 utterances in total, of which 16,397 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each feature's 'relevant'-class F1-score alone vs. experiment 3 (↑ higher, ↓ lower, → similar).* 42

4.21 Leave-one-feature-out ablation on the test set: macro-averaged metrics. *Note: the test dataset contains 24,375 utterances in total, of which 16,397 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: change in macro F1-score when each feature is removed, vs. experiment 3 (↑ improvement, ↓ decline, → negligible).* 43

4.22 Leave-one-feature-out ablation on the test set: relevant-class metrics. *Note: the test dataset contains 24,375 utterances in total, of which 16,397 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: change in the 'relevant'-class F1-score when each feature is removed, vs. experiment 3 (↑ improvement, ↓ decline, → negligible).* 43
4.23 Emotion polarity coefficients and their impact on the SVM model. . . . 45
4.24 Agreement and disagreement among the VADER, human, and LLM emotion labels on the 50 samples. 47
4.25 Agreement and disagreement between human validation, the 'gold', and the full SVM model on the 50 sampled utterances. 48
4.26 The seven utterances where human validation and the 'gold' disagree; the samples marked from 2 to 7 have summaries that draw on a wider context than the utterance itself. 48
4.27 Sampled utterances whose extracted persona includes information not stated in the utterance (18 of the 50 cases). 50
4.28 Per-class and macro-averaged precision, recall, F1-score, and support for the three relevance-prediction scenarios. 51

B.1 The 50 randomly sampled utterances with their human validation, 'gold' relevance (extracted persona), SVM predictions, and VADER, human, and LLM emotion labels. 71

Chapter 1

Introduction

Dialogue is an important component of human interaction. Understanding how humans interact, the frequency of certain words, what they remember or forget from past interactions, the emotions conveyed by particular words, and how they express themselves can reveal important aspects of how people relate to each other in conversation.

The rapid advance of large language models (LLMs) has raised questions about how to personalize the experience for users, and for this to happen an LLM must be able to 'remember' aspects of those users. The current research focuses on how to build long-term memory for LLMs Wu et al. (2025) that can extract and retain information, and decide what is relevant about a user's persona that is worth keeping across multiple conversational sessions. LLMs have become remarkably useful conversational assistants (Brown et al. (2020)); however, they still struggle with long-term memory, especially when deciding what is relevant to remember in a conversation with a human.

In a long-term system, the focus is on identifying the utterances that contain important and relevant information that should be remembered. In a multi-turn conversation, not every utterance has the same value: some turns introduce valuable information about the persona facts, habits, preferences, while others are greetings, filler, or simply interactional conventions of human dialogue. Relevance therefore becomes a central concept in the study of memory in dialogue for long-term interaction. Yet this concept is hard to define and has been called a 'fuzzy' term, since it can mean different things depending on the context and the people taking part in the conversation.

To investigate relevance in human dialogue, we begin by looking at the MSC dataset created by Facebook AI, and in particular at the MSC:PersonaSummary subset. MSC is a multi-session dialogue dataset in which speakers continue their conversation after a temporal gap, so that common ground has to be updated and rebuilt across sessions. This means that some information is refreshed and reused while other information is forgotten, depending on its importance and on how the conversation shifts. When a conversation is resumed after a temporal gap, as in a natural setting, only some information remains accessible and relevant for future interactions. The MSC dataset was created to mimic long-term conversation that unfolds over time, where the interlocutors re-engage in the discussion, bringing in information from the past or introducing new topics.

1.1 Research Questions

The present study is guided by the following central questions:

What linguistic cues in human-human interaction indicate what is 'relevant' for continuing a conversation over the long term?

Can we use these linguistic cues to promote more human-like multi-turn conversations in human-LLM interaction?

In order to answer them, we begin by reviewing several linguistic theories that can shed light on the cues that may signal relevance. These theories center on relevance theory, common ground and grounding, discourse markers, and the given-new contract. The study focuses on identifying the linguistic cues that may mark a relevant utterance, one that contains an important persona aspect. The cues we consider important include first-person references, possessive phrases, discourse markers, negation, temporal expressions, emotion polarity, interjections, and position in the dialogue. These insights can then be leveraged to improve the long-term memory of LLM systems and, as a result, promote more personalized multi-turn conversations between LLMs and humans.

Based on these cues, which we consider important aspects of human-human dialogue, we develop a feature engineering approach that helps us identify the utterances containing relevant persona facts. We chose an SVM model in order to offer a transparent and interpretable view of how relevance is selected. We compare several feature representations and train using a linear SVM classifier to classify each utterance as relevant or not. The relevant utterances are then selected as candidate memory items and used directly as extractive summaries. The goal is not to summarize everything, but to identify which information is reused later and is therefore relevant for long-term dialogue.

Next, we conduct a feature ablation study to assess the impact of each feature. We assess the quality of the output using BERTScore and LLM as a judge to complement the SVM model experiments. We then carry out a manual error analysis on a subset of samples to understand some of the system's errors. The error analysis includes a human evaluation on this subset, compared against the 'gold' in the data and against the SVM model's predictions on relevance.

In the final step, we use an LLM to indicate which information is relevant and not at the utterance level. This is again compared with the SVM model and with the crowd worker annotations of what they consider relevant to keep in a multi-turn conversation. The goal of the thesis is to understand which aspects of human dialogue indicate what information is important to retain as relevant and which can be allowed to fade naturally.

Chapter 2

Related Work and Linguistic Theories

2.1 Related Work

Over the years, numerous datasets have been collected for use in open-domain dialogue research, with an emphasis on long-term memory for dialogue systems. These datasets were based either on natural conversations or on crowdsourced conversations. LLMs handle memory differently from humans (Liu et al., 2023), specifically 'humans can keep thoughts in the memory and recall them without repeated reasoning', while LLMs use 'repeated recall-reason steps', i.e. LLMs repeatedly reconstruct the information and reason from scratch every time. As a result, it is important to understand what is relevant in human-human dialogue in order to relate it to the way LLMs generate responses and how human-LLM interaction can be improved. Another important aspect is understanding why memory is important in human-human interaction.

In early works, different approaches were explored. Zhang et al. (2018) noted that chit-chat models lack several important aspects, such as specificity, consistent personality traits, and engaging conversational behavior. In the paper *Personalizing Dialogue Agents: I Have a Dog, Do You Have Pets Too?* Zhang et al. (2018) introduced the dataset PERSONA-CHAT, in which each crowd worker was asked to engage in a conversation while simulating an assigned persona. The pairs were instructed to engage in natural conversation in order to get to know each other. The resulting conversations were engaging, and the systems aimed to mimic this style of interaction. The experiments showed that persona-related information improved the model's prediction of subsequent utterances in conversation (Zhang et al., 2018).

Mazaré et al. (2018) argued that dialogue systems face an important limitation: they often fail to engage users effectively in conversation. In *Training Millions of Personalized Dialogue Agents*, Mazaré et al. (2018) introduced a dataset containing five million personas extracted from Reddit and concluded that training models on diverse personas improved their performance. This confirmed the importance of persona-conditioned conversational agents and the role of information about the participants involved in the conversation in understanding dialogue quality.

Xu et al. (2022) moved beyond single-session persona-based dialogue and created Multi-Session Chat (MSC), a human-human dataset. In *Beyond Goldfish Memory: Long-Term Open-Domain Conversation*, the authors highlighted that long-term memory has not been sufficiently studied and that models are trained and evaluated on short

conversations, which provide limited context. This dataset contains multiple chat sessions in which speakers try to get to know each other and discuss each other's interests as well as topics raised in earlier sessions. The authors designed the dataset to capture how conversation continues in real life when speakers resume interaction after hours or days.

To address the memory issue, Xu et al. (2022) explored several strategies, including transformer encoder-decoder fine-tuning, which is limited by token length; retrieval augmentation, which improves generation by retrieving relevant information from past conversations; and summarization-based memory augmentation, which summarizes past conversations and stores them. Rather than increasing the context window of transformers, the authors suggest retrieval and summarization as more effective strategies.

However, these systems still have limitations (Xu et al., 2022): retrieval-based methods can incorporate historical data into the current context, but they still struggle with coherence. In many studies, the focus has been on how these systems remember past information using summarization methods, without taking into consideration that some information needs to be forgotten and updated as it naturally occurs in human-human dialogue. It was assumed that once a system learned a fact, it remained permanently true and relevant, but this is not always the case, because in real-life situations topics and preferences can change on a daily basis. The systems must be able to maintain a constant flow of information and retain only what is relevant and important in chronological order. If a system is unable to retrieve and update the most recent information, the conversation may contain contradictory memories, which will be unhelpful for users. Our interpretation is that such dialogue may be perceived as incoherent.

Bae et al. (2022) pointed out that users' situations change constantly in real life. Before long-term memory systems were developed, stored information was treated as permanently valid, and earlier literature did not take into consideration that users' preferences often changed. Bae et al. (2022) argued that memory is dynamic and noted that memorized information can become outdated at some point. In this paper, the authors proposed a method to address this issue through the following operations: PASS, REPLACE, APPEND, and DELETE. These operations do not only store information; they also eliminate outdated or invalid information and update past information. This prevents the system from retaining irrelevant history that conflicts with the user's current situation.

However, long-term memory in conversation is not only about storing information from past interactions; it is also about identifying relevance. Jang et al. (2023) indicated that the time interval and the relationship between speakers can be important for relevance in dialogue. Jang et al. (2023) introduced the Conversation Chronicle dataset, arguing that varying time intervals (from hours to years) and relationships between speakers shape the evolution of contextual information across multi-session dialogues. However, they do not analyze the linguistic aspects that occur in this kind of real-world conversation because they rely on LLMs to create and simulate human interaction.

In a more recent work, several approaches were presented to address the incoherence of these systems. Wang et al. (2025) proposed a method through which LLMs generate and update dialogue summaries automatically. This allows the model to maintain consistent responses in long-term conversations. The authors argued that because the generated summaries are shorter than the full dialogue, the proposed memory schema also offers a solution for current LLMs by helping them handle long-context dialogue sessions without changing or increasing the context-length settings (Wang et al., 2025,

p. 4).

Liu et al. (2023) argued that humans do not recall the full history of a conversation; instead, they recall the thoughts, deductions, and impressions formed during interaction. In order to replicate this mechanism, they introduced the Think-in-Memory (TMI) method. This framework helps systems update and evolve thought through operations such as insertion, forgetting, and merging (Liu et al., 2023). TMI helps model LLMs in a more human-like way, where the memory mechanism is more about remembering and selecting thoughts in a long-term interaction, while engaging in a 'post-thinking' operation that allows LLMs to merge information or forget some user-related information. The shift is important, and Liu et al. (2023) argued for moving from data storage toward a concept more closely related to memory and thought storage.

However, these papers do not identify what is relevant in human dialogue. Our research here can provide a better understanding of what is important for humans to keep and what to forget, which can be very important in the interaction between humans and LLMs.

Most of the existing work mainly focuses on storing and retrieving past information, while less attention has been paid to how relevance and discourse-level mechanisms shape what is retained, reused, or forgotten in human dialogue. For this reason, linguistic and pragmatic theories are needed to understand how meaning is negotiated across time. Relevance theory, common ground, discourse markers, and given-new contract provide a framework for examining this process.

2.2 Linguistic and Pragmatic Perspectives

Qamar et al. (2025) argue that LLMs have the capacity to understand dialogue. Although the performance of LLMs in text generation is high, the question is whether they truly understand the meaning of dialogue or whether they are merely 'stochastic parrots' (Bender et al., 2021) that mimic dialogue without understanding the meaning behind the words. Qamar et al. (2025) argue that understanding the broad context of an utterance involves more than understanding its semantic content, because even if two utterances have the same semantic content, it does not mean that they convey the same meaning (Qamar et al., 2025, p. 26219). Meanings differ depending on the context and on what the speaker wants to convey. LLMs have a limited capacity to understand conversational pragmatics. Qamar et al. (2025) demonstrated through the DA (Dialogue Acts) classification task that LLMs have difficulty understanding the underlying dynamics of conversation. Through the pre-tasks, such as Turn Management, Communicative Function Identification, and Dialogue Structure, the authors found that LLMs struggle with simple fundamental tasks that are very easy for humans to complete (Qamar et al., 2025, p. 262220). It was highlighted that LLMs lack comprehension ability when it comes to dialogues. The analysis presented by Qamar et al. (2025) showed that LLMs struggle to identify the turn-taking roles of speakers and that, in most cases, they rely on the nearest utterance for context. They fail to capture the dependency relation between utterances if they are not close to each other (Qamar et al., 2025, p. 262220). They also demonstrated that LLMs are biased toward viewing utterances as having a backward-looking function even when that is not the case, e.g. when a new topic is introduced.

2.3 Relevance and Common Ground

Our main idea is that relevance in dialogue does not refer to lexical similarity, but rather to how the participants in the dialogue are able to infer meaning through context and shared assumptions while reaching a mutual common ground. Theories of relevance and common ground are helpful in defining a conceptual basis for understanding what information is relevant and important to keep in the context of long-term memory. It is not only about retaining information; the focus should be on which information can be used, reactivated, or reused across multiple temporal gaps. When an interlocutor goes back to a conversation after minutes, hours, or days, the events from the past conversation are not only about how the data is extracted or about the act of extraction, but more about the social interaction and negotiation needed to reach a certain common ground. Grice in *Logic and Conversation* introduced the maxim of relation, 'Be relevant' (Grice, 1975, p. 46), which argues that the speakers need to make their contribution relevant for the ongoing conversation and when this expectation is not fulfilled, the listeners deduce the information through *implicature* (Bender and Lascarides (2019)). It means that the listeners interpret the information that was suggested but wasn't explicitly expressed. The interlocutors rely on previously established and shared facts in order for the assumptions to be reliable.

In this way, some of the cognitive aspects of human interaction can be revealed and this can provide a framework showing which aspects govern relevance in human-human conversation. Through the lenses of the theories of relevance (Sperber and Wilson, 1995) and common ground (Clark and Brennan, 1991), we can form hypotheses that we can make predictions about what linguistic cues may mark relevance and whether this can explain why LLMs fail to perform this complex social negotiation.

2.3.1 Relevance Theory

Relevance is an important aspect of human interaction and communication. Sperber and Wilson (1995) sought to clarify the concept of relevance. The authors argue that they are not trying to define the word 'relevance', because this term has different meanings for different people and is therefore called a 'fuzzy term' (Sperber and Wilson, 1995, p. 119). According to Sperber and Wilson (1995), human communication is based on a mental process aimed at achieving optimal relevance. Because it is a mental process, it implies 'a certain effort, a certain expenditure of energy' (Sperber and Wilson, 1995, p. 124). Relevance is determined through two aspects: cognitive effect, which refers to the usefulness of the context for a certain piece of information, and processing effort, which refers to the mental effort and energy required to interpret the message. It shapes the way participants in a conversation deduce the meaning of utterances and how they access previous knowledge from the contextual information they already have about a certain topic. This also implies a process of deduction used to access stored contextual knowledge from previous information, and 'each context (apart from the initial context) contains one or more smaller contexts, and each context (apart from the maximal contexts) is contained in one or more larger contexts' (Sperber and Wilson, 1995, p. 142). Speakers produce utterances with the expectation of relevance, while the hearer's task is to use the contextual information to understand the intended interpretation. This theory places relevance at the center of comprehension, with utterances produced against past knowledge and the other interlocutor searching for an interpretation that yields cognitive effects worth the effort (Sperber and Wilson, 1995).

This theory is important for the current thesis because it provides one main insight: the relevant information in a dialogue is not necessarily the most recent information, but the information that holds meaningful aspects worth storing and that can help in understanding human interaction. However, the older relevant information needs to compete with the recent one for the retrieval process, and it must be marked linguistically.

In a long-term dialogue that occurs after hours, days, or weeks, the information that contributes may concern personal facts, preferences, or references used later to support utterances and maintain continuity in the conversation. Sperber and Wilson (1995) also suggest a connection between relevance and memory. Relevance theory suggests that memory is not just a passive mechanism through which information is stored, but an organized system inseparable from the concept of relevance. The authors argue that 'the way in which information is chunked may in principle help or hinder the search for relevance; plausibly, forms of chunking which are a help rather than a hindrance tend to predominate' (Sperber and Wilson, 1995, p. 150). This suggests that the way information from a conversation is 'chunked' in memory has a major impact on whether the search for relevance is facilitated or obstructed. They also highlight the importance of the 'organisation of encyclopaedic memory' (Sperber and Wilson, 1995, p. 150) in the search for relevance. The notion of human 'encyclopaedic memory' suggests the capacity to organize interactions into 'chunks' that help minimize future effort in processing information. This can show that LLMs store information as a relatively flat memory and that the 'chunks' are not always organized according to the relevance of a given conversation, which can make their responses seem disconnected, unnatural, or robotic when they bring up past information. From the perspective of relevance theory, we may find an explanation for why LLMs sometimes struggle to retrieve and bring up memories that are relevant to the actual context. Because of their inability to organize past facts that match the conversational context, LLMs and their retrieval methods may not always align with the principles of relevance that govern human memory.

2.3.2 Common ground

To complement relevance theory, we introduce the notion of common ground and grounding theory. Clark and Brennan highlight that communication and interaction are collective activities that depend on common ground, which can be defined as mutual knowledge, beliefs, or assumptions. They consider these aspects to be 'collective actions,' through which the participants in a conversation must constantly update their common ground (Clark and Brennan, 1991, p. 127). The authors formulated a definition for grounding, which states that 'in communication, common ground cannot be properly updated without a process we shall call grounding' (Clark and Brennan, 1991, p. 128). They argue that, in conversation, participants try to establish whether the information provided was understood. These aspects are important in dialogue because they reveal that a successful conversation is not only about producing grammatically and linguistically correct utterances, but also about sustaining coordination between interlocutors over time. Clark and Brennan emphasize that interlocutors seek evidence that the other understood the message and that they are on the same path, and this can be done through acknowledgment, repairs, and similar signals. In a multi-session context, this involves retrieving the right information from a past interaction as an attempt to access and reuse the personal common ground between the participants.

An utterance or piece of information is considered relevant and grounded when both participants in the conversation acknowledge it. Information becomes relevant when it can be retained as common ground and used in future interactions.

The authors also discuss the 'cost of grounding,' which includes the cost of formulation and production (the effort of speaking or typing), the cost of reception and understanding (the effort of comprehending the message), and the repair cost that occurs when a misunderstanding needs to be fixed (Clark and Brennan, 1991, pp. 142, 145). These are related to the principle of 'least collaborative effort' (Clark and Brennan, 1991, p. 134), through which participants in dialogue try to minimize the conversational work required to provide and understand information in order to achieve mutual understanding. Clark and Brennan argue that 'collaborative effort' changes significantly depending on the medium and on physical constraints. In face-to-face conversation, people can express understanding or misunderstanding through facial expressions, but in a text-messaging medium this is not available, and it lacks facial or auditory grounding as well as vocal prosody. In such cases, interlocutors must rely on explicit lexical cues, such as discourse markers, to navigate temporal gaps and update common ground.

Relevance theory and common ground together provide a research direction for long-term dialogue. Relevance theory helps explain why some utterances matter more than others and which ones are relevant to keep and reuse, while grounding explains the way meanings are shared and updated across interactions.

2.4 Discourse markers as linguistic cues

Along with relevance theory and common ground, discourse markers are a useful linguistic resource for identifying what is relevant in dialogue. Discourse markers help identify how one utterance relates to another and they provide a way to discover how human interaction is structured. In *Discourse Markers*, Schiffrin defines these linguistic cues as 'sequentially dependent elements which bracket units of talk' (Schiffrin, 1987, p. 31). Rather than being mere conversational fillers without much significance, these markers play an important role in discourse: they help guide the conversation and allow interlocutors to manage the flow of information and navigate memory retrieval. Discourse markers such as 'well, but, and, so, you know, I mean, now, and then' contribute much more maintain the information flow than propositional content. With them, the flow of conversation is organized, and they guide interlocutors in interpreting what is being said. We can say that discourse markers function as contextual cues that connect relevant information in dialogue. They also help maintain topical coherence and allow speakers to continue the conversation, shift to a new topic, or bring up information from past interactions, which can be interpreted as temporal progression. Schiffrin considered that discourse markers are used on different planes of talk: exchange structures, action structures, idea structures, participation frameworks, and information states (Schiffrin, 1987, p. 315), and that markers often establish more than one contextual coordinate at once (Schiffrin, 1987, p. 330). The author also mentioned that there are four dimensions of context (speaker, hearer, time, and place) (Schiffrin, 1987, p. 322), which are often encoded through deictic elements such as personal pronouns, temporal expressions, including tense, and locative expressions, including verbs of motion (Schiffrin, 1987, p. 322).

These are relevant in long-term dialogue when interlocutors must reconnect new

utterances with earlier parts of the conversation and maintain continuity across the dialogue while restoring temporal gaps. Because of this, discourse markers are important linguistic cues that help identify which information is useful to keep active and relevant in ongoing dialogue over time, updating or re-establishing common ground. Different discourse markers serve as linguistic cues that indicate contrast, continuation, reformulation, paraphrase, coherence, and temporal progression.

2.4.1 Given-New Contract Theory

The presented aspects of discourse markers, which function as linguistic cues signaling new information, can be linked to the given-new contract theory (Clark and Haviland, 1977). Through this theory, Clark and Haviland provide a way to understand how information is organized in dialogue. The main idea is that interlocutors do not present ideas as a sequence of facts; rather, they structure them around what is new and what is given, where the given information is already shared in previous discourse and comes from a shared context in the past. Shared knowledge is a node in the listener's memory structure, and the listener stores the content of the conversation in long-term memory (Clark and Haviland, 1977, pp. 4-5). According to the authors, the given-new strategy has three steps: at Step 1, the listener isolates the given and the new information in the current sentence. At Step 2, the listener searches memory for a direct antecedent, a structure containing propositions that match the given information precisely. Finally, at Step 3, the listener integrates the new information into the memory structure by attaching it to the antecedent found in Step 2 (Clark and Haviland, 1977, p. 5).

If one of the interlocutors fails to establish the given context, the listener must construct a link in order to understand the relevance of the sudden shift in the flow of the conversation. Clark and Haviland call this 'bridging' (Clark and Haviland, 1977, p. 6), which requires high cognitive effort. Through this theory, we can assume that comprehension in a conversation becomes easier when the listener is able to connect new information to an already shared and existing knowledge structure, which allows the conversation to flow naturally and maintain continuity over time. For this reason, the given-new contract theory can be placed alongside relevance theory and common ground theory: information is easier to understand and process in a conversation when it can be associated with what the interlocutors already know and share.

This theory concludes that listeners rely on the given-new strategy to interpret sentences. When told something, they compute what is given and what is new in the utterance, search memory for an antecedent of the given information, and then add the new information to memory (Clark and Haviland, 1977, p. 38). This theory is important for understanding why some information is easier to reuse and more relevant than other information. In a natural multi-session conversation, interlocutors often refer to aspects of earlier topics, such as personal facts or shared experiences and knowledge, which can be categorized as 'given' in later interactions. If this information was already shared or established, then new utterances are easier to understand, and the 'new' or unfamiliar information needs to be processed in order to be added to the common ground. The authors mention this in the following way: 'Sometimes it is impossible to find any way of bridging the gap between known information and the appropriate antecedent. Then the listener must add to memory, perhaps hypothetically, a new node (a nominal associated with one or more propositions) to serve as the antecedent to the given information' (Clark and Haviland, 1977, p. 7).

In the context of long-term memory retrieval, we believe that discourse markers can

be associated with the given-new contract as signaling factors. By inserting markers such as ‘you know’ into the conversation, the speaker establishes the ‘given’ information and prepares the listener to receive the ‘new’ information, updating the shared knowledge and the common ground.

Chapter 3

Dataset Exploration and Analysis

3.1 Dataset Description

The Multi-Session Chat (MSC) **(Data Statement in Appendix B)** dataset was developed by Facebook AI. It consists of dialogues between humans over five sessions, with each session having up to 14 utterances. Previously, conversational datasets were created as single, isolated conversations, while MSC was created to mimic the natural conversations that take place over time between the same people. The purpose of this dataset was to conduct research on long-term conversation (Xu et al., 2022). The sessions are 'paused' (Xu et al., 2022), and the speakers resume or continue the chat after a simulated amount of time, such as hours or days.

The crowd workers were asked to continue the chat about the previous topic, bring up shared subjects from the past, or engage in new topics.[1] They were asked to play a provided role that contained some of the characteristics of that role. The authors mentioned that the dataset was divided into three parts: training, containing 4,000 episodes with three sessions and 1,001 episodes with four sessions; validation, extended up to five sessions; and test, also extended up to five sessions (Xu et al., 2022). During these interactions, the interlocutors are instructed to learn things about each other.

This dataset is useful for this research because the research questions focus on what is relevant in long-term conversation, memory, and which information is relevant enough to be reused in the continuity of an interaction. For this reason, MSC:PersonaSummary was used because it contains the conversations and the extracted persona facts which are used by interlocutors to create summary for each session. This dataset processes the summaries as a long-term memory mechanism and consists of user traits, past events, shared knowledge, and experiences. The selection of this subset from the dataset followed several factors: from a linguistic perspective, the summaries contain the interlocutors' shared knowledge and common ground, and they can be related to the theory of the Given-New Contract. Given information is old information from the previous sessions, which is carried over the next session following temporary gaps.

This dataset is supported in ParlAI (Miller et al. (2018)) for conversational exploration and while ParlAI is great for exploring conversations in a streaming manner, it's not well suited for bulk data analysis. For this reason, we decided to use the JSON files published for this dataset directly and then saved them in the CSV format for easier

[1]They were also asked not to share any private or sensitive information about themselves. The purpose was to protect privacy and to 'ensure diversity even if the same crowd worker conducts multiple conversations' (Xu et al., 2022).

analysis and inspection. All the further analysis was then performed using the pandas Python library Wes Mckinney (2010) and the derivative features were also saved in CSV for further analysis.

We created two CSV datasets out of the MSC:PersonaSummary for the thesis, one is for all the utterances and corresponding extracted persona and another dataset for all sessions. The utterance dataset has the columns described in table 3.1 and 3.2.

Table 3.1: Utterance Dataset Columns

Column Name	**Description**	**Example**
`initial_data_id`	Unique identifier representing the episode id between two same interlocutors.	`test_0`
`session_id`	Number representing the specific chat session.	`1`
`turn_index`	Sequential order of the message within the conversation, starting at 0.	`0`, `1`, `2`
`speaker`	Speaker who spoke this utterance in the conversation.	`bot_0`, `bot_1`
`text`	Actual dialogue or utterance spoken by the speaker in that turn.	"What kind of car..."
`text_word_count`	The total number of words contained within the `text` column for that specific turn.	`11`
`extracted_persona`	Specific persona facts relevant for future conversation extracted from the text in *that single turn.*	"I own a Jeep."
`known_personas`	A cumulative list of all the background facts gathered about that speaker up to that point in the conversation, including the initial persona facts given to each speaker at the start of the episode.	`['I own a Jeep.', ...]`

Table 3.2: Session Dataset Columns

Column Name	Description	Example
initial_data_id	Unique identifier representing the episode id between two same interlocutors.	test_0
session_id	Number representing the specific chat session.	1
time_elapsed_num	Amount of time passed since last session.	3
time_elapsed_unit	Unit for the time passed.	days
total_turns	Total number of messages in the session.	15
speaker_1_init_personas	Starting facts known about speaker 1.	['I love dogs.']
speaker_2_init_personas	Starting facts known about speaker 2.	['I am a nurse.']
speaker_1_session_summary	Summary of speaker 1's relevant personsa facts in the chat.	"Discussed new pets."
speaker_2_session_summary	Summary of speaker 2's relevant persona facts in the chat.	"Talked about work."
speaker_1_followup_question	Opening question speaker 1 can ask next time.	"How is the puppy?"
speaker_1_new_fact_for_next_session	New persona trait added for the next chat.	"Adopting a rescue."

3.2 Dataset Statistics and Analysis

We provide more precise statistics on the conversational part of MSC:PersonaSummary to find linguistic cues that can show what is relevant in dialogue between humans. We present the dataset from a macro point of view before moving to micro observations of the dataset.

3.2.1 General Statistics

We started with general session and utterance level statistics that indicated the number of unique episodes, the total number of sessions, the total number of turns, the average number of tokens per utterance, the total number of utterances per session and how many of those utterances contributed to summaries (extracted persona) per session, and the frequency of how many sessions contain a given number of turns. This is repeated for all datasets - training, validation and test. The tables 3.3, 3.4 and 3.5 show the statistics for all datasets.

Table 3.3: General session-level and utterance-level statistics across dataset splits.

Metric	Training	Validation	Test
Total Unique Episodes	4,000	500	501
Total Sessions	10,285	2,000	2,004
Total Turns/Summaries	133,290	25,459	24,375
Avg Tokens per Utterance	18.10	20.02	21.01

Table 3.4: Distribution of turns per session across dataset splits.

Turns in a Session	Number of Sessions		
	Training	Validation	Test
6	4	–	–
7	103	–	–
8	130	36	26
9	142	30	23
10	220	25	26
11	101	25	8
12	6,043	1,384	1,653
14	1,332	138	260
16	2,113	346	6
18	58	12	1
20	19	3	–
22	4	–	1
24	4	1	–
26	5	–	–
28	2	–	–
30	5	–	–
Total	**10,285**	**2,000**	**2,004**

Table 3.5: Per-session utterance counts and summary extraction across dataset splits.

Dataset	Session ID	Utterances	With Summary		No Summary	
			Count	%	Count	%
Training	1	59,894	36,370	60.72	23,524	39.28
	2	46,420	29,646	63.86	16,774	36.14
	3	26,976	17,559	65.09	9,417	34.91
Validation	1	7,768	5,111	65.80	2,657	34.20
	2	5,897	3,808	64.58	2,089	35.42
	3	5,890	3,718	63.12	2,172	36.88
	4	5,904	3,795	64.28	2,109	35.72
Test	1	6,572	4,658	70.88	1,914	29.12
	2	5,939	3,825	64.40	2,114	35.60
	3	5,924	3,903	65.88	2,021	34.12
	4	5,940	4,011	67.52	1,929	32.48

A few observations: the training dataset has more episodes than the validation and test sets; training dataset has 3 sessions with summaries each episode while the

validation and test sets have 4 sessions with summaries per episode. The training dataset has longer sessions compared to validation and test. Overall, each session from train, validation, and test has more than 60% of utterances with extracted persona contributing to summary.

3.2.2 Macro Statistics - Training Dataset

Training Dataset

The next step was to tokenize the part of the dataset that contains the conversations in order to find out the number of unique words in the dataset. First, punctuation marks and stop words were not removed, and the total number of unique words was 36,236. The most frequent words were: *'.', 'I', 'to', 'you', ',', 'a', '?', 'the', '!', 'and', 'it'*, etc. Second, punctuation marks and stop words were removed, and the total number of unique words was 33,280. The most frequent words were the following: *('like', 21,681), ('love', 12,680), ('good', 10,880), ('get', 10,575), ('would', 10,565), ('really', 9,086), ('time', 8,653), ('work', 8,248), ('great', 7,887), ('go', 7,711).* This signals that the dialogues are oriented toward positive reactions, evaluations, and preferences. The tokens also give insight into the topics the crowd workers discussed, and they are oriented toward daily routines, which is compatible with persona-based interactions and fits into long-term dialogue by providing subjects for the continuity of the conversation in the future.

The next step was to measure lexical diversity by calculating the type-token ratio in two cases, before and after removing the punctuation marks and stop words. In the first case, the type-token ratio was *0.01* and in the second case it was *0.03*. After filtering the data, the type-token ratio is higher, which is to be expected because the punctuation markers were quite frequent before the data was refined and were reducing lexical diversity. This also indicates that humans generally use repetitive vocabulary, which can be connected with the principles of Relevance Theory. In comparison with other dialogue datasets reported in the (Krause, 2025), where the average type-token ratio was 0.06, the type-token ratio for our current dialogue dataset is lower. The lower type-token ratio indicates more repetition and this suggests that interlocutors narrow the size of the vocabulary, which is more efficient for expressing patterns of their personas and for providing information related to their daily routines, preferences, etc., while reducing the effort of the interlocutors.

Following the same pattern for the extracted summaries from the sessions, the results were: *total unique tokens - 13,651; type-token ratio - 0.02; total unique tokens after filtering - 8,245; type-token ratio after filtering - 0.04*. In the summaries, words such as *'like', 'love', 'work',* etc. were also among the most frequent. The type-token ratio shows a more diverse vocabulary than in the raw conversation, because the information in the summaries is condensed, and only the most important and relevant information is kept.

The dataset contains an average utterance length of *18.09* and an average of *33.32* utterances per episode. This signal that the interlocutors are engaged in constructing personas and sharing information that can be used as Given information in the next sessions.

The distribution of sentences per utterance indicates that most utterances are short, with one to three sentences being the most common pattern, while longer utterances occur rarely. The total number of sentences in all utterances was *289,847*. The numbers

are shown in the table 3.6.

Table 3.6: Distribution of sentence counts, per utterance (raw conversations) and per session summary.

Number of Sentences	Utterances	Session Summaries
1	40,072	7,367
2	51,322	6,210
3	27,401	3,882
4	9,776	1,903
5	3,235	782
6	1,005	288
7	322	84
8	90	29
9	41	6
10	17	13
11	6	6
12	1	–
13	1	–
15	1	–

Next, we measured the average number of tokens per utterance with and without filtering the data. Average number of tokens per utterance without filtering was *21.21*, and with filtering was *8.33*.

Afterwards, we checked the number of sentences in all the summaries, which is *45,753*. The average number of tokens per summary before filtering was *9.06*, and after filtering was *2.82*. The summaries are shorter than the raw conversations, with the most frequent summaries containing between one and three sentences. This indicates that they are concise and compress the factual information from the dialogues. The table 3.6 shows the frequency of sentences in summaries.

Analyzing the distribution of content words and function words in the dialogue is useful to understand the linguistic structure of the dataset and which information is conveyed. Based on the categorization described in Tools for Clear Speech (2026), we categorized these two categories using POS tags in the following way: content words: {*'NN', 'NNS', 'NNP', 'VB', 'VBD', 'VBG', 'VBN', 'VBP', 'VBZ', 'JJ', 'JJR', 'JJS', 'RB'*}; function words: {*'IN', 'DT', 'PRP', 'CC', 'TO', 'MD', 'WDT', 'WRB', 'PRP$', 'UH'*}. The automated extraction of POS tags using the SpaCy library Honnibal et al. (2020) indicates that the total number of content tokens in the dialogue was 1,392,030 and the total number of function tokens was 1,026,107 respectively. Punctuation was also removed, since punctuation marks do not have a POS category in this classification. For the summaries, the counts were 384,533 for content tokens and 238,824 for the function tokens respectively.

Content words are the words that carry meaning, such as places, names, hobbies, personal facts, etc., and they are important for memory and relevance in long-term dialogue. Function words are the words that link ideas together, and they are useful for understanding how the dialogue is structured and held together, and how coherence is maintained. Even if function words do not carry factual data, they are still essential in a conversation for managing the flow of conversation over temporal gaps. The number of content words being higher than the number of function words is expected because

content words carry the semantic information from the dataset and because they belong to an open class.

MSC dataset is informationally dense because the crowd workers were instructed to find out information about each other, and they were engaged in persona discovery that takes place in multiple sessions over temporal gaps. In this dataset, many utterances contain information about personal facts and preferences that are related to the given persona, activities, references to past sessions, and shared knowledge, which increases the frequency of nouns, adjectives and verbs. By contrast, the number of function words still demonstrates the human effort and dedication used in typed conversation to have a coherent and structured flow of interaction, the main purpose being the organization of discourse. The difference demonstrates that the dataset is rich in content and is informationally oriented.

Next, we found the most frequent POS tags (extracted using the SpaCy library Honnibal et al. (2020)) in the conversation part and in the summary part. For the conversations, the most frequent POS tags were: *('PRP', 333,009), ('NN', 297,588), ('.', 264,574), ('IN', 213,123), ('RB', 182,220), ('DT', 180,772), ('VBP', 176,726), ('JJ', 167,744), ('VB', 164,012), ('NNS', 91,913).* For the summaries, the most frequent POS tags were: *('NN', 116,162), ('PRP', 108,250), ('VBP', 99,010), (',', 75,801), ('.', 45,957), ('DT', 45,386), ('IN', 41,056), ('JJ', 34,934), ('NNP', 29,332), ('VB', 22,867).*

In the conversation data, the predominant POS tags were 'PRP' and 'NN'. Pronouns dominate the dialogue, which is different from the summaries, where the predominant POS tag is the noun. This may be because the raw conversation follows a natural pattern that happens in daily conversations, where the interlocutors often refer to themselves and to the other interlocutor, creating an interactive exchange of information. This can establish common ground. In the summaries, the interactional dimension is reduced, and the focus is on semantic content, such as preferences, people, and personal facts. The summaries reflect what is relevant to keep from the conversation. The dialogues introduce the information, while the summaries retain what is relevant, a shift that can be interpreted through the linguistic theories presented earlier: Relevance Theory, common ground, and the Given-New Contract.

An exploration of the time elapsed between sessions indicates that it does not affect the continuity of the dialogue in any meaningful way. The assumption that the interlocutors would share more information when the time between two sessions was longer didn't hold in statistics as the amount of time was injected randomly and they do not show genuine natural behavior. The statistics are in the table 3.7.

Table 3.7: Average number of turns by simulated time gap between sessions.

Hours Between Sessions	Average Number of Turns
168	11.64
144	11.64
120	11.60
96	11.65
72	11.67
48	11.77
24	11.72
7	11.68
6	11.73
5	11.66
4	11.65
3	11.68
2	11.69
1	11.71

These statistics were representative for the macro representation of the MSC:PersonaSummary dataset. In the next section, we present some of the findings from a micro point of view.

3.2.3 Micro (Linguistics) Statistics - Training Dataset

Training Dataset

In spoken conversation, interlocutors rely on body movements and facial expressions to understand if the other person understood the Given information and if the common ground was updated with the new information. In typed conversation, however, these techniques are not available, and interlocutors rely on other factors, such as discourse markers. In this part, we analyze the summaries instead of the the raw conversation because the dialogues contain conversational behavior, while the summaries represent condensed information extracted from the dialogues.

Discourse Markers

The discourse markers were extracted from the utterances using a phrase matcher for specific discourse lexicons in the SpaCy library. The analysis shows that a small number of discourse markers appear frequently, especially *'well', 'actually', 'you know', 'I guess', 'anyway', and 'I mean'*. These markers are used to indicate hesitation, reformulation, to organize the response, and to connect responses with other utterances. Relating them to Relevance Theory, they can signal a reduction of effort in communicating information and can help interlocutors connect new information to a prior memory, discourse, or utterance. Another important factor is that they help in updating the common ground, or they make the understanding of shared information more visible and maintain the continuity of discourse. In the following list, we present the 20 most frequent markers: *('well', 6,028), ('kind of', 2,234), ('actually', 1,912), ('you know', 1,274), ('I guess', 1,007), ('anyway', 283), ('honestly', 259), ('sort of', 248), ('I suppose', 192), ('I mean', 188), ('to be honest', 181), ('well, actually', 158), ('well,*

you know', 146), ('pretty much', 143), ('basically', 125), ('well, kind of', 99), ('well, I guess', 98), ('literally', 57), ('kind of, actually', 44).

We can notice the dominance of *'well'*, which, according to Schiffrin in *Discourse Markers*, has the primary function of anchoring the speaker as a respondent in the interaction (Schiffrin, 1987, p. 316). It is used to maintain cooperation between the interlocutors, and in different situations it can indicate a pivot in shifting the topic. It also gives the speaker a moment to evaluate what was said before initiating their own clarification or perspective. In the MSC dataset, the interlocutors are required to retrieve information from past interactions across temporal gaps, and 'well' helps them maintain the floor by introducing a delay in the interaction that is required for fulfilling the Given-New Contract and searching long-term memory in order to bridge the context with new information and utterances.

'You know' and *'I mean'* are taken together in Schiffrin's study. According to Schiffrin, *'y'know'* is a marker that has the intention of obtaining attention from the hearer in order to open a conversation whose purpose is to provide information, while the goal of *'I mean'* is to maintain the focus on the speaker (Schiffrin, 1987, p. 267).

Schiffrin argues that 'Y'know is used to create a situation in which the speaker knows about (has meta-knowledge of) knowledge which is shared with the hearer' (Schiffrin, 1987, p. 269), while 'I mean' indicates the interlocutor's intention to modify the meaning or ideas presented in a prior utterance (Schiffrin, 1987, p. 296) and 'maintains speaker focus on prior material. But note that *I mean* also maintains hearer focus on prior material: *I mean* instructs the hearer to continue attending to the material of prior text in order to hear how it will be modified' (Schiffrin, 1987, p. 309). The author concluded that 'y'know' marks the intention in transition in the shared knowledge that is common with the interlocutor, while 'I mean' indicates the focus of the speaker on his or her own talk (Schiffrin, 1987, p. 309).

Also, 'you know' in some situations has a complementary relationship with the interjection 'oh', which appears in the raw dialogue 3,564 times, along with emotional states such as 'oh, wow' 505 times and 'oh, no' 410 times. Schiffrin suggested that it contributes to interactional relevance by marking a reaction from the speaker that indicates the relevance of the utterance. 'Oh' is used to express emotional states such as surprise, fear, and even acknowledgment. It is relevant for common ground because 'oh' can indicate that the interlocutor is providing an emotional reaction to new information and helps in updating shared knowledge. In long-term dialogue, these markers offer support in helping the interlocutors connect new information with earlier utterances, which is useful for the continuity of the dialogue.

In the statistics presented above, we can also see markers that can be considered epistemic hedging, such as 'kind of' - 2,234 times, and 'I guess' - 1,007 times. These signal uncertainty. Some other markers, such as 'anyway', can indicate continuation of the topic or a shift in conversation, while 'actually' indicates clarification or correction of information and could be considered a mechanism for repairing the common ground, allowing the interlocutor to update the grounding without interrupting the flow of the dialogue. All these markers signal that memory retrieved by humans can be imperfect and suggest that interlocutors may misinterpret shared knowledge. In some cases, the interlocutors try to navigate through the conversation while simultaneously taking time to look into long-term memory ('well') and correct a fact or show their uncertainty with markers such as 'well, actually' - 158 times, and 'well, I guess' - 98 times.

Interjections

Beyond these markers, the statistics on interjections reveal that in most utterances these tokens are not used, but the most frequent ones were used in acknowledgment: 'yes' - 5,701 times, 'yeah' - 3,886 times, and 'sure' - 359 times. They were used for confirming and understanding, and for maintaining common ground, being considered a form of positive evidence and called back-channel responses (Clark and Brennan, 1991). In conversation, they are important because they validate that new information was accepted, or that the interlocutor agrees with the information retrieved from shared knowledge. Jefferson (1984) calls 'yeah' an 'acknowledgement token', alongside 'mmmm' and 'uh'. In this study, 'yeah' is a token that indicates speakership incipiency: a shift from being a listener to being an active speaker and taking the floor. Drummond and Hopper (1993), in *Back Channels Revisited: Acknowledgment Tokens and Speakership Incipiency*, consider 'yeah' an active token. We can infer from this that the interlocutor is being active in updating the common ground or accepting the new information that was provided.

In addition to being a discourse marker, 'Well' also appears in the interjection statistics 2,303 times, which indicates the double functionality of this token in the dialogue. As an interjection, it can signal sudden emotions.

We also notice another group of tokens that express positive evaluation and affective responses, such as 'cool', 'nice', 'lol', and 'haha'. This indicates that the conversations flow in a natural way by maintaining cooperative relationships across temporal gaps.

Emotion

Going forward with the micro-analysis of the raw conversation, we noticed that most of the emotions expressed (calculated using VADER score - Hutto and Gilbert, 2014) are neutral, appearing 87,432 times (65.60%), which is expected because the dataset is created to share personal facts and preferences. It is interesting to notice that the strong emotions expressed in the interactions are positive compared with the negative ones, appearing 43,518 times (32.65%) and 2,340 times (1.75%) respectively. The positive emotions by far exceed the negative ones, which might be predictable in a dataset in which the crowd workers are asked to play a role. This can also signal the desire of humans to be polite in interactions with others. This characteristic can be interpreted through Politeness Theory. This theory is presented in *Politeness: Some Universals in Language Usage* by Penelope Brown and Stephen C. Levinson. 'Positive face' expresses 'the positive consistent self-image or 'personality' (crucially including the desire that this self-image be appreciated and approved of) claimed by interactants' (Brown and Levinson, 1987, p. 311). The theory argues that people generally try to maintain face in their interactions with others. These numbers suggest that the dataset is socially oriented, and from the perspective of long-term dialogue and common ground, it signals a constant positive interactional tone that assures the continuity of the conversation across multiple sessions.

Agency Phrases

In the next part of the statistics, the focus was on extracting how many cases include constructions with the first person ('I') and possessive phrases. In 70% of cases, the utterances contain constructions driven by 'I', which can signal the rather subjective nature of the dataset. In this context, however, the participants constantly provide

information about themselves and their preferences while trying to establish and update the Given baseline of the interactions. Also, in 22.4% of the utterances, the crowd workers used possessive phrases, which can suggest that they provided information about ownership, actions, personal capabilities, and intentions. Even if the percentage is not high, it is still significant because it expresses that in many utterances, personal facts were encoded and can be used in the context of long-term memory.

Next, we define agency phrases as clauses where a specific person or group ('I', 'me', 'myself', 'we', 'us', 'ourselves') acts as the direct subject or object of a verb. It signals the instances where the speaker explicitly associated themselves with an action. We extracted these using the SpaCy library. We find out how many utterances that contain agency phrases also contain strong positive or negative emotions. Agency phrases were used in 58,144 neutral cases (66.50% out of all neutral cases), 1,925 strong negative cases (82.26% out of all strong negative cases), and 33,993 strong positive cases (78.11% out of all strong positive cases).

These patterns can be used to signal that the dataset contains factual information and interactional components that can be considered relevant in maintaining the common ground and the shared knowledge that can be updated constantly across the sessions.

Temporal Deictics

We calculated the statistics to find out the frequency of 'then' and 'now' as temporal deictics. In Schiffrin's study on discourse markers, these two tokens are considered time deictics affecting the direction of discourse: 'now points forward in discourse time and then points backward. 'now' focuses on how the speaker's own discourse follows the speaker's own prior talk; 'then', on the other hand, focuses on how the speaker's discourse follows either party's prior talk' (Schiffrin, 1987, p. 246). It is difficult to extract these two tokens since they are polysemous words: they can be temporal adverbs or discourse markers. This means that their meaning changes depending on the context of an utterance. They are relevant for long-term memory and the continuity of conversation, helping speakers organize events and reconnect earlier utterances with new ones. POS extraction and the dependency parser with specific constraints on verb tenses in the SpaCy library helped find these temporal tokens, however, they may have still overestimated such token use. With these limitations, we found 256 utterances that used temporal 'then' and 2570 utterances used 'now'. Most of the utterances where 'then' and 'now' were used contributed to the extracted summary, pointing to a strong correlation with relevance.

Negation

Negation is an important aspect of conversation, and it can signal the denial of a piece of information or the exclusion of something from the persona profile. It can also be connected with updating the common ground. Negation has been described as a 'contrary speech-act' (Givón, 1978), in which one interlocutor denies the beliefs of the other participant in the conversation. In some contexts, a non-event can serve as an important piece of information; such non-events are salient because they may highlight relevant aspects of a conversation.

Even though most people use positive sentences in conversation and negative sentences may be rare, those negative sentences can contain information worth retain-

ing in memory for future conversations. Negation assertions 'are used when - much more rarely in communication - one establishes the event rather than stasis as ground' (statis refers to non-event) (Givón, 1978). We extracted negation patterns on an event or action using dependency matching with the SpaCy library, matching the instances where an action verb (main or auxiliary) has dependent child labeled as negation. In MSC:PersonaSummary, the raw conversation contains 24,326 utterances with negation out of 133,290 utterances, which means that 18.25% of utterances have negation patterns. This is consistent with Givón's theory that negation statements are less frequent in conversation than affirmative ones.

Even though the percentage is small compared with affirmative sentences, negation can still indicate relevant information. The most frequent negation particle is 'n't', which occurs in 51.75% of the negation cases, followed by 'not' at 34.60%. This signals that users negate actions and states of being, since these particles are associated with the verb in most cases. The table 3.8 gives an overview.

Table 3.8: Distribution of the negation particles within the negated utterances.

Negation word	**Percentage (%)**
n't	51.76
not	34.61
never	11.16
nt	2.44
no	0.03
nope	0.003

From a pragmatic perspective, these cues are used to express preferences, reject assumptions, and update the common ground. In long-term conversation, they are relevant because they indicate dislikes, exclusions, or the denial of certain experiences.

Temporal Expressions

Another set of linguistic cues that can indicate relevance in an utterance are temporal expressions, also called time deixis (Levinson, 1983). These can be considered supportive cues that point to references to past sessions, recency, and future plans, and that ensure continuity across time. They also indicate how the interlocutors organize memory and conversation around temporal cues. In Levinson's theory, time deixis is context-dependent, and these expressions include tokens such as today, tomorrow, yesterday, last week, and this week. Levinson notes that 'it is important to distinguish the moment of utterance (or inscription) or coding time (or CT) from the moment of reception or receiving time (or RT)' (Levinson, 1983, p. 73). The CT refers to the moment the message was produced, while the RT refers to the moment it was read by the interlocutor. Temporal expressions depend on when an utterance was produced and when it was received, and some of them must be interpreted according to the time gap between the sessions, since one of the interlocutor may recall an utterance from previous session as memory.

Based on this, the next step was to check the temporal state of the sentences, whether they indicate recency, past, or future. The percentages are shown in the table 3.9.

Table 3.9: Distribution of temporal states across the utterances.

Temporal state	**Count**	**Proportion (%)**
STATE_DEFAULT_PRESENT	120,173	90.16
STATE_RECENCY	6,108	4.58
STATE_PAST	4,380	3.29
STATE_FUTURE	2,629	1.97

This shows that most utterances are treated as present by default (90.16% of the cases), and only small percentages are classified as recency, past, or future. This suggests that most utterances center on the current interaction. The rest of the data contains explicit time deixis indicating recency (4.58%), past (3.29%), and future (1.97%). Even though these are minority cases, they are important for the relevance of the dialogue.

Among the extracted temporal tokens, the most dominant is 'now' (30.78%). This may reflect the interlocutors' tendency to anchor the conversation in the immediate present. To refer to past history, they often use 'then' and 'once' (14.71% and 8.49%, respectively), which shift the coding time backwards, while 'soon' (10.76% of the cases) looks forward to the future, perhaps to plans. Even so, the token 'then' can be tricky, as it does not always indicate the past, given its complexity. The phrase 'used to' appears in 7.01% of the cases and can indicate past habits. The table 3.10 gives an overview of these temporal deictics.

Table 3.10: Frequency of the extracted temporal-deixis tokens.

Word	**Count**	**Proportion (%)**
now	4,037	30.78
then	1,930	14.71
soon	1,411	10.76
once	1,114	8.49
recently	1,112	8.48
used to	920	7.01
currently	462	3.52
later	412	3.14
Now	343	2.61
tomorrow	325	2.48
someday	277	2.11
Then	242	1.84
Once	151	1.15
eventually	130	0.99
nowadays	71	0.54
Currently	47	0.36
Someday	24	0.18
Recently	23	0.18
Soon	15	0.11
Tomorrow	15	0.11
Used to	12	0.09
Eventually	12	0.09
previously	9	0.07
Later	8	0.06
Nowadays	7	0.05
presently	4	0.03
ONCE	1	0.01
NOW	1	0.01
Presently	1	0.01
Previously	1	0.01

In the table, we also notice that tokens such as ‘now’, ‘then’, and ‘later’ appear capitalized as well. We decided to keep them as they appear, because depending on the context they may play a different syntactic role in the sentence, with emphasis on the timeline or on the action, and ‘now’ and ‘then’ can function either as time deixis or as discourse markers. Their interpretation requires analysis of the surrounding context.

Next, named-entity extraction (computed directly with the SpaCy library) shows the presence of temporal expressions such as ‘today’ (2,194 times, 10.69%), ‘tonight’ (604 times, 2.94%), and ‘this weekend’ (508 times, 2.47%). This signals that the interlocutors anchor their topics and utterances in time, making references to the past, present, and future. These aspects are important in long-term conversation and memory because they connect the conversation and its events with earlier actions.

3.2.4 Proportions Across Sessions - Training Dataset

Training Dataset

Computing the proportions for each session lets us see how things evolve across sessions. These proportions are summarized in the following tables 3.11 and 3.12.

Table 3.11: Per-session statistics (part 1 of 2). 'Total turns' is a count; 'mean word count' is an average; the remaining columns are percentages.

Session ID	Total turns	Mean word count	Mean Extracted facts	Mean Negation rate	Mean Interjection count
1	59,894	10.52	60.72	10.68	27.29
2	46,420	23.67	63.86	23.65	28.18
3	26,976	25.32	65.09	25.78	27.42

Table 3.12: Per-session statistics (part 2 of 2). All values are percentages.

Session ID	Mean Discourse marker count	Mean Agency count	Mean Possession count	Mean Strong emotion polarity	Mean Temporal state change count
1	6.75	64.72	20.61	24.74	5.20
2	14.84	74.82	24.26	41.56	13.53
3	16.03	76.23	23.21	43.56	13.79

We notice that even though the number of turns is higher in session 1, the mean number of words is lower than in the following sessions, where it roughly doubles (10.52, 23.67, and 25.32, respectively). This may signal that many of the turns in the first session were greetings and introductory chat, building the routine for future conversation. It may also indicate that the interlocutors are strangers creating the environment in which they will begin to share relevant information about their personas. We also notice that, although the volume of utterances decreases from session 1 to session 3, the information density is higher. This suggests that sessions 2 and 3 are more about the 'anchor position' (Schegloff, 1986, p. 116) that follows once the interlocutors have exchanged most of the 'how are you' greetings and begin to drop in updates about their personas.

The proportions of extracted facts are similar across the sessions (60.72%, 63.86%, and 65.09%, respectively), even though there are fewer utterances in sessions 2 and 3. The negation rate and the discourse-marker rate are more than double in sessions 2 and 3 compared with session 1. This may mean that in session 1 the interlocutors are simply presenting some facts about themselves, and that by session 2, once the conversation has evolved, they are updating the common ground and revising their beliefs about the past interactions. They do this by using discourse markers to link the new information with the given information, while negation is used for denial and for correcting outdated information. The percentage of interjections is similar across the sessions. Agency phrases (64.72%, 74.82%, and 76.23%) and possessive phrases (20.61%, 24.26%, and

23.21%) are also similar across the three sessions, with small differences. This still indicates that, as the conversation evolves, the dialogue becomes more self-centered, providing personal information - facts, ownership, actions, intentions, and capabilities.

These patterns indicate that, for relevance in long-term conversation, later sessions rely on information centered on persona consistency and on shared and updated memory and grounding.

We also notice differences across sessions for strong emotion polarity (from 24.74% to 43.56%) and temporal state change (from 5.20% to 13.79%). This may signal that the exchanges in the first session were more neutral and static, and that this changes as the conversation evolves. The interlocutors also rely heavily on temporal deixis to express their life updates chronologically in order to share the common ground. This may signal that, in long-term interactions, later sessions depend on shared history and temporal expressions to anchor life updates in the current conversation.

Chapter 4

Methodology and Experiments

4.1 Feature Description

Based on the presented linguistic theories and the dataset exploration, we have identified some of the indicators that can signal relevance within the dialogue in the table below. We will engineer them as features for building the model to identify relevance in utterances from the multi-turn conversation dataset.

Table 4.1: Overview of the linguistic cues and features and their theoretical justification.

Linguistic cues / Features	**Theoretical justification**
First-person references	Provide persona facts that can be relevant in long-term conversation.
Discourse markers	Signal given-new information and help guide the flow of information and memory retrieval.
Possessive phrases	Support common ground and provide information about ownership, intentions, and personal capabilities.
Emotion polarity	Strong positive: reflects politeness theory, which can ensure the continuity of the conversation. Neutral: indicates factual information and routines. Strong negative: indicates a sad situation or negative life event, or expresses clear dislike or frustration.
Interjections	Signal engagement in the conversation, acknowledgment, and reactions to information.
Negation	Can signal correction, dislikes, and the denial of previous assumptions, and helps update common ground.
Temporal expressions	Locate information in the past, present, or future across sessions.
Position in the dialogue	Includes the session stage and the turn stage.

4.2 Hypothesis

Based on the linguistic theories and our analysis of the dataset, we formulated the following hypotheses. **H1**: Natural human dialogue often contains filler words, discourse markers, interjections, first-person references, negation, possessive phrases, temporal expressions, emotion polarity, and similar cues. These cues may indicate information that is relevant to a conversation and that can be reused later in the interactions. In the dataset, utterances that contain these cues are more likely to be relevant and retained in the summaries than utterances that contain few such cues or none at all. These linguistic cues help ground shared information.

H2: By relying on conversational mechanisms and not on raw text alone, a model can indicate which utterances are more likely to be relevant for memory. Feature engineering can serve as a method for combining linguistic cues with lexical information to identify relevance, rather than relying on lexical information alone.

H3: Interpretable modeling methods such as SVM offer reasonably good performance and transparency for retaining information for memory in multi-turn conversation.

4.3 Modeling

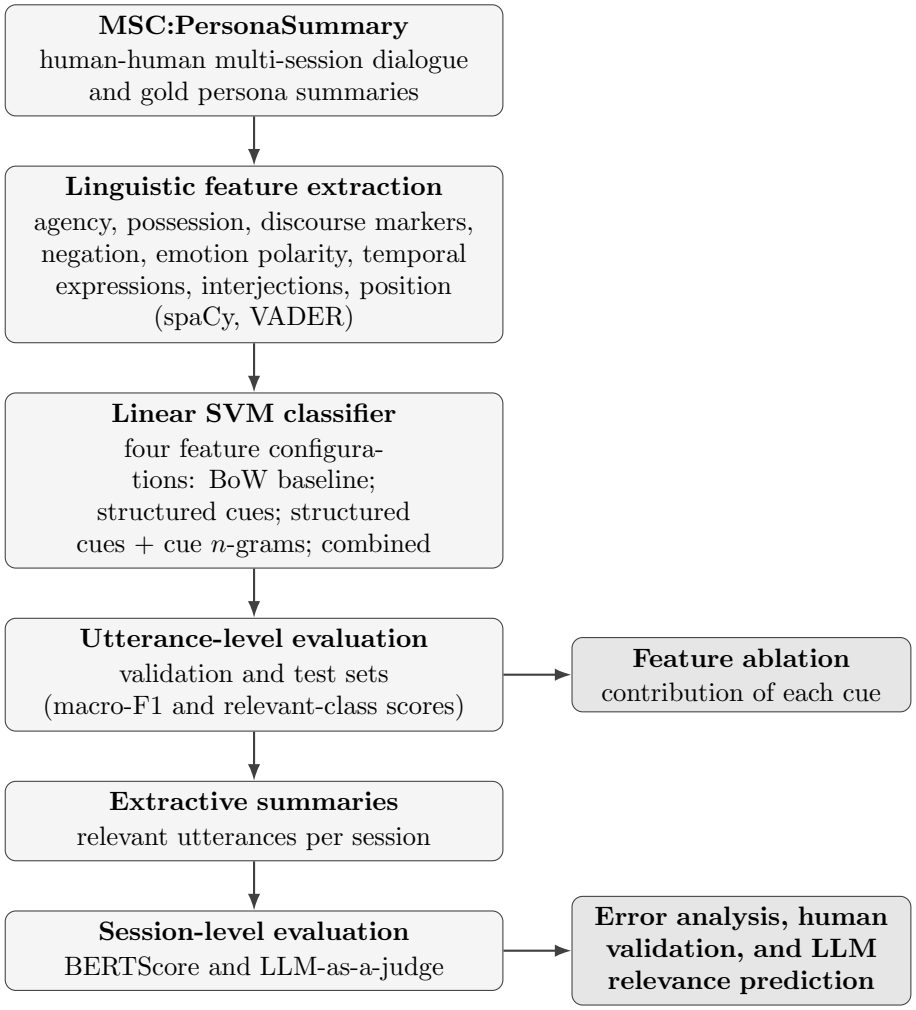


Figure 4.1: Overview of the methodology

In the figure 4.1, we give an overview of the methodology steps starting with the MSC:PersonaSummary dataset, followed by extracting linguistic features and feature engineering, training linear SVM classifier, evaluation at utterance-level and session-level, feature ablation, and closing with the error-analysis and LLM-based relevance check.

The overall experimental design follows a supervised learning pipeline using a linear Support Vector Machine (SVM) with different combinations of linguistic and lexical

features to evaluate their contribution to utterance-level relevance detection. The model is first evaluated at the utterance level using gold-standard labels on the validation dataset. This is followed by a session-level evaluation of the generated summaries using BERTScore and an LLM-as-a-judge approach. A feature ablation study is then conducted on the validation set to assess the impact of individual feature groups. Once the experimental setup is finalized, the same pipeline is applied to the test dataset. Finally, an error analysis is performed on a randomly selected subset of 50 utterances from the test dataset, followed by a human evaluation of the same sample to compare them against the gold labels.

The Table 4.2 is an example of the utterance dataset created out of MSC:PersonaSummary and Table 4.3 is an example of the features extracted.

Table 4.2: MSC Utterances Dataset Example

initial_data_id	**session_id**	**turn_index**	**speaker**	**text**	**text_word_count**	**extracted_persona**	**known_personas**
test_0	1	0	bot_0	What kind of car do you own? I have a jeep.	11	I own a Jeep.	['I own a Jeep.']
test_0	1	1	bot_1	I don't own my own car! I actually really enjoying walking and running, but then again, I live in a small town and semi-close to work.	26	I live semi-close towkr. I don't own a car. I enjoy running and walking. I live in a small town.	['I live semi-close towkr.', "I don't own a car.", 'I enjoy running and walking.', 'I live in a small town.']
test_0	1	2	bot_0	Ah I see! I like going to the gym to work out.	12	I enjoy exercising at the gym.	['I own a Jeep.', 'I enjoy exercising at the gym.']

Table 4.3: Extracted Features Example

Feature	**Turn 0**	**Turn 1**	**Turn 2**
initial_data_id	test_0	test_0	test_0
session_id	1	1	1
agency_phrase	i have	i enjoying, i live, i own	i like, i see
possession_phrase		my car	
interjection			ah
discourse_marker	kind of	actually	
named_entities			
negation		n't	
emotion_polarity	neutral	neutral	neutral
emotion_score	0.0	0.3284	0.3612
emotion_words	kind	enjoying	like
temporal_state	STATE_ DEFAULT_ PRESENT	STATE_PAST	STATE_ DEFAULT_ PRESENT
trigger_word		then	

Following the hypotheses on linguistic cues and feature engineering, we adopted a supervised machine learning approach using SVM. The SVM model used for the experiments is linear, and the system's settings were kept at their default values. The aim is to test whether linguistic cues related to common ground, relevance, dialogue structure, and given-new information can help identify utterances that are relevant and that contain persona facts important for future sessions.

The purpose is not to rely on raw conversation and vocabulary alone, but to engineer features that connect the machine learning model with dialogue structure and human conversational pragmatics. In these experiments, we create a binary **gold label** (relevant/not relevant) based on the 'extracted persona' annotated column. 'extracted persona' column indicates which persona fact was extracted from each utterance and signals a practical proxy for relevant information that can be reused in long-term conversation and memory. We consider an utterance as relevant if a gold persona fact was extracted from it by the annotators.

To assess the impact of linguistic cues on how humans decide what is relevant to retain for future sessions, we conducted four experiments. We used Bag-of-Words (BOW) and one-hot vector representations of the utterances to represent the features. The features are: 'agency phrases', 'possession phrases', 'interjections', 'discourse markers', 'negation', 'emotion polarity' (emotion words used as an auxiliary feature), and 'temporal state' (trigger words used as an auxiliary feature), as well as 'session stage' and 'turn stage'. The feature 'session stage' has three categories based on the session index: 'early' (session 1), 'middle' (session 2), and 'late' (session 3 and later). The feature 'turn stage' similarly has three categories, but these are based on the position of the turn within the session: 'early' for the first third of turns, 'middle' for the second third, and 'late' for the final third. The experiments aim to determine whether an utterance

is 'not relevant' or 'relevant'.

The first experiment is a baseline that only uses the text of the utterances, vectorized with Bag-of-Words (BOW) to represent the vocabulary they contain. It offers insight into whether simple word frequency alone is sufficient to extract the persona facts that can be reused or updated as common ground in future sessions.

The second experiment uses one-hot vectors that combine presence of linguistic features (binary indicators of whether a feature is present) with categorical features. The binary features marking presence are 'agency phrases', 'possession phrases', 'interjections', 'discourse markers', and 'negation', while the categorical features are 'emotion polarity', 'temporal state', 'position' ('session stage', and 'turn stage'). This experiment relies on the system's linguistic feature representation rather than on the raw text of the utterances, and it indicates the presence or absence of linguistic cues.

The third experiment combines the second experiment with linguistic text features represented as n-grams. Here, the linguistic cue markers are concatenated as n-grams, so that the representation captures not only the presence of a cue (for example, a discourse marker) but also the specific words used. The n-grams are restricted to sizes 1 to 4 to limit the vocabulary size and the resulting features are represented as one-hot vectors. This is intended to help the SVM model capture local linguistic patterns while avoiding an overly large and noisy vocabulary. The experiment tests whether combining the presence of linguistic cues with the actual words improves the identification of relevance in human dialogue.

The fourth experiment is a hybrid approach that combines experiments 1 and 3 (BOW on text utterances + one-hot vectors + n-grams). It brings together the text-based representations and the linguistic features in order to examine how the SVM model performs relative to using each representation on its own. By merging these feature types, the model can weigh the meaning of an utterance while simultaneously taking its linguistic cues into account, which helps determine whether the utterance contains relevant information that can be reused in future interactions.

4.4 Utterance Level Evaluation - Validation Dataset

For the preliminary results, the SVM model was trained using scikit-learn (Pedregosa et al. (2011)) on the training dataset and evaluated at the utterance level on the validation dataset. The validation dataset contains a total of 25,459 utterances, of which 16,432 hold relevant information and 9,027 hold not relevant information. Because the two classes are imbalanced, with relevant utterances making up roughly two thirds of the data, overall accuracy can be misleading; we therefore use the macro-average F1-score as the primary metric, which weighs both classes equally and report the per-class scores along with it. In tables 4.4 and 4.5), in each column, the best value across the experiments is shown in bold, and the arrows indicate the change relative to the previous experiment as the experiments build on each other by adding more features.

Table 4.4: Utterance-level results overview on the validation set (macro-averaged).

Experiment	Gold relevant utterances	Predicted utterances	Accuracy	Precision (macro)	Recall (macro)	F1-score (macro)
Experiment 1 (bow_baseline)	16,432	15,900	0.76	0.74	0.75	0.74
Experiment 2 (structured_cues)	16,432	18,945	0.77 ↑	**0.76** ↑	0.72 ↓	0.73 ↓
Experiment 3 (structured_cues _plus_cue_ngrams)	16,432	15,796	0.77 →	0.75 ↓	0.75 ↑	0.75 ↑
Experiment 4 (combined_ full_model)	16,432	15,534	**0.78** ↑	**0.76** ↑	**0.77** ↑	**0.76** ↑

Table 4.5: Per-class utterance-level results on the validation set.

Experiment	Class	Precision	Recall	F1-score
Experiment 1 (bow_baseline)	not_relevant	0.66	0.69	0.67
	relevant	0.83	0.8	0.81
Experiment 2 (structured_cues)	not_relevant	**0.75** ↑	0.54 ↓	0.63 ↓
	relevant	0.78 ↓	**0.9** ↑	**0.84** ↑
Experiment 3 (structured_cues_plus_cue_ngrams)	not_relevant	0.66 ↓	0.71 ↑	0.68 ↑
	relevant	0.83 ↑	0.8 ↓	0.82 ↓
Experiment 4 (combined_full_model)	not_relevant	0.67 ↑	**0.73** ↑	**0.7** ↑
	relevant	**0.85** ↑	0.8 →	0.82 →

In the table 4.4, we see for the **first experiment (bow baseline)**, the macro-average precision, recall, and F1-score are 0.74, 0.75, and 0.74 respectively. Because the focus of this thesis is to identify relevance in human dialogue, it is also worth examining the scores for the relevant class are 0.83, 0.80, and 0.81 for precision, recall, and F1-score respectively. A more detailed classification report for this experiment is shown in Figure A.1. This experiment shows that raw vocabulary, based on word frequency, can be useful for capturing most of the factual information in the conversation.

The **second experiment (structured cues)** predicted more utterances as relevant than gold which suggests that the system is more permissive when deciding whether an utterance contains relevant information, and that it also marks some not relevant utterances as relevant. The macro-average precision, recall, and F1-score are 0.76, 0.72, and 0.73 respectively and for the relevant class, they are 0.78, 0.90, and 0.84. The full classification report for this experiment is shown in Figure A.2. Here, the high recall for the relevant class comes at the cost of lower precision. Because the model relies only on the presence of linguistic cues and the categorical features, without any vocabulary, it over-predicts the relevant class.

In the **third experiment (structured cues plus cue ngrams)**, the macro-average precision, recall, and F1-score are 0.75, 0.75, and 0.75 respectively and for the relevant class, they are 0.83, 0.80, and 0.82. The full classification report for this experiment is shown in Figure A.3. This suggests that the surface cue expressions encoded as n-grams add useful information and stops the model from over-predicting compared to Experiment 2.

In the **fourth experiment (combined full model)** the macro-average precision, recall, and F1-score are 0.76, 0.77, and 0.76 respectively, giving this system the highest

macro-average F1-score of all four. It also achieves the highest precision for the relevant class (0.85), with recall and F1-score of 0.80 and 0.82, respectively. Although it predicts fewer relevant utterances than the other three experiments, it appears to best optimize the trade-off between the not relevant and relevant classes. The full classification report for this experiment is shown in Figure A.4. This experiment shows that combining vocabulary with the linguistic cues raises the precision for the relevant class to 85%, a small but consistent gain over the bag-of-words baseline (83%). It favors quality over quantity, gaining this precision by filtering out false positives.

Overall, the fourth experiment outperforms the other three. This suggests that combining linguistic cues, lexical information, and human conversation rules is an important direction for identifying relevance in multi-turn conversation.

4.5 Utterance Level Evaluation - Test Dataset

In this part, we report the results on the test dataset. Because the original paper does not specify exactly how the data was split, we ran the experiments and report results for both the validation and test sets. We observe similar patterns across the two, which suggests that the results are stable. The test dataset contains a total of 24,375 utterances, of which 16,397 hold relevant information and 7,978 hold not relevant information. The test results are close to those on the validation dataset, with only small differences. Here too, experiment four has the highest macro-average F1-score, as well as the highest precision for the relevant class.

The next tables 4.6 and 4.7 give an overview of the experiments from the macro-average and binary perspectives for the test dataset (as before, the best value in each column is in bold, and the arrows show the change from the previous experiment).

Table 4.6: Utterance-level results overview on the test set (macro-averaged).

Experiment	**Gold relevant utterances**	**Predicted utterances**	**Accuracy**	**Precision (macro)**	**Recall (macro)**	**F1-score (macro)**
Experiment 1 (bow_baseline)	16,397	15,677	0.77	0.74	0.75	0.74
Experiment 2 (structured_cues)	16,397	18,315	**0.78** ↑	**0.76** ↑	0.72 ↓	0.73 ↓
Experiment 3 (structured_cues_plus_cue_ngrams)	16,397	15,104	0.76 ↓	0.73 ↓	0.75 ↑	0.74 ↑
Experiment 4 (combined_full_model)	16,397	15,242	0.77 ↑	0.75 ↑	**0.76** ↑	**0.75** ↑

Table 4.7: Per-class utterance-level results on the test set.

Experiment	Class	Precision	Recall	F1-score
Experiment 1 (bow_baseline)	not_relevant	0.63	0.69	0.66
	relevant	0.84	0.8	0.82
Experiment 2 (structured_cues)	not_relevant	**0.71** ↑	0.54 ↓	0.61 ↓
	relevant	0.8 ↓	**0.89** ↑	**0.84** ↑
Experiment 3 (structured_cues_plus_cue_ngrams)	not_relevant	0.61 ↓	0.71 ↑	0.66 ↑
	relevant	0.85 ↑	0.78 ↓	0.81 ↓
Experiment 4 (combined_full_model)	not_relevant	0.63 ↑	**0.73** ↑	**0.68** ↑
	relevant	**0.86** ↑	0.8 ↑	0.83 ↑

In the **first experiment (bow baseline)**, the macro-average precision, recall, and F1-score are 0.74, 0.75, and 0.74 respectively. The scores for the relevant class are close to those on the validation dataset, at 0.84, 0.80, and 0.82 for precision, recall, and F1-score. The full classification report for this experiment is shown in Figure A.5. The results are again close to those on the validation dataset, and this experiment shows once more that raw vocabulary, based on word frequency, can be useful for detecting the majority of the factual information.

In the **second experiment (structured cues)**,as with the validation dataset, the experiment over-predicts what is relevant in the dialogue. Because the model relies only on the presence of linguistic cues and the categorical features, without any vocabulary, it over-predicts the relevant class at the expense of the not relevant class. The macro-average precision, recall, and F1-score are 0.76, 0.72, and 0.73 respectively, while for the relevant class they are 0.80, 0.89, and 0.84. The full classification report for this experiment is shown in Figure A.6.As in the validation dataset, the high recall for the relevant class again comes at the cost of lower precision.

The **third experiment (structured cues plus cue ngrams)** produces results similar to the results on the validation dataset. The macro-average precision, recall, and F1-score are 0.73, 0.75, and 0.74, respectively, while for the relevant class they are 0.85, 0.78, and 0.81. The full classification report for this experiment is shown in Figure A.7. The hybrid experiment again shows that including the explicit text of the linguistic cues as trigger words gives the model the context it needs to filter out false positives. The n-gram sizes also help guide the model in capturing lexical patterns.

The **fourth experiment (combined full model)** once again shows that combining the linguistic cues with vocabulary gives the highest precision for the relevant class (86%), a small but consistent improvement over the bag-of-words baseline (84%). The full classification report for this experiment is shown in Figure A.8.

As a reference point, a majority baseline that labels every utterance as 'relevant' has an accuracy of about 0.65 on the validation set and 0.67 on the test set, mainly because most utterances are relevant, yet it scores a macro-F1 of only about 0.39 (validation) and 0.40 (test), since it fails completely on the 'not relevant' class. Our best model (experiment 4 - combined full model) reaches a macro-F1 of 0.76 on validation and 0.75 on test, an improvement of roughly 0.35 over this majority-class baseline. As we will see later in the single feature ablation study, when used on their own, features such as position, negation, temporal and discourse markers collapse to this floor (macro-F1 of about 0.39), because in isolation they simply label almost every utterance as relevant.

Overall, these results confirm the pattern observed on the validation dataset: combining lexical information with the linguistic cues derived from linguistic theory yields

robust performance on noisy, unseen data. The minimal differences between the validation and test sets indicate that the model generalizes well within this distribution. This consistency is compatible with the model capturing regularities associated with the Given-New contract rather than fitting to a particular lexical distribution, although it should be noted that the strongest configuration is the one that also includes the lexical bag-of-words features. Taken together, the results indicate that lexical information and theory-based linguistic cues are complementary and respond well to unseen data.

4.6 Session Level Evaluation - Validation and Test Dataset

Next, to evaluate the methodology on the validation dataset at a session level, we compare the summaries compiled from the relevant utterances marked by the model with the gold summary created by combining the persona facts for each session. We use BERTScore and an LLM as a judge for scoring in this set of evaluations. We use the summaries from experiment 4 (combined_full_model), because it has the highest scores among the experiments.

4.6.1 LLM as a Judge

For the LLM as a judge we use prompt engineering. The goal at this step is to measure the factual coverage of the persona, that is, to check whether the model flags the utterances that contain and preserve the persona facts present in the gold summary, such as habits, preferences, relationships, personal information and important statements. The model used for this purpose is **Gemini 3.1 Pro Preview**, accessed through **Google AI Studio**. To make the LLM judge's evaluation as deterministic as possible, we set the temperature to 0.0, the thinking level to high and the top-p to 0.1. The model was given system instructions to be strict about factual coverage and to ignore conversational filler and differences in wording. The LLM assessed the quality of each generated session summary in three steps: the first two steps identify the factual statements in both gold summary and model generated summary, count them and then compare the facts from model summary with the 'gold' summaries to see how many were captured; the third step computes the coverage (recall = captured facts / gold facts) and precision score (captured facts / total model facts). We also defined the score scale for recall: 0 when none of the 'gold' facts are captured and 1 when all of them are, and for precision: 0 when there were no model facts. For each session summary produced by our model, the LLM judge computes the recall and precision, and returns a score between 0 and 1 (**LLM prompts for summary session evaluation in Appendix B**).

To feed the validation dataset to the LLM, we had to split it into batches because of the length of the data and the token limit. The validation dataset contains 500 episodes, each with 4 sessions, for a total of 2,000 sessions. The sessions were fed to the LLM in four batches of 500 sessions, one session per row. For each session summary, the LLM returned a score between 0 and 1. We used the experiment 4 summaries, which also include the text utterances. The results for each batch were collected in a CSV file, and each coverage score was computed as captured facts / total gold facts.

The coverage (recall) scores by session are in table 4.8: session 1, 0.825; session 2, 0.782; session 3, 0.777; session 4, 0.782. The mean coverage for the validation dataset is 0.792. These scores follow the recall metric. A mean coverage of roughly

80% indicates that the SVM-based pipeline is able to extract a substantial amount of long-term information.

We then computed precision over the same batches for a full picture of the summaries, as captured facts / total model facts. The precision scores are: session 1, 0.802; session 2, 0.761; session 3, 0.756; session 4, 0.751. The mean precision for the validation dataset is 0.768, giving an F1-score of 0.778. In both cases, session 1 scores higher than the others. The next table 4.8 reports precision, recall and F1-score by session, together with the mean.

Table 4.8: Session-level LLM-as-a-judge scores on the validation set. *Arrows compare each session with the overall mean (↑ above, ↓ below).*

Metric Level	**Sessions**	**Precision Score**	**Recall Score**
Session-Level	↑ 1	**0.802**	**0.825**
	↓ 2	0.761	0.782
	↓ 3	0.756	0.777
	↓ 4	0.751	0.782
Validation Dataset	Overall Mean	**0.768**	**0.792**
	Overall F1 Score	**0.778**	

The LLM judge shows that the summaries generated by the SVM model have strong factual coverage and good factual precision. The model captures a good share of the gold persona facts while including only a limited amount of information absent from the 'gold' annotated summaries.

To check that the extraction pipeline does not tend to overfit, we ran the same process on the test dataset, with the results in table 4.9.

Table 4.9: Session-level LLM-as-a-judge scores on the test set. *Arrows compare each session with the overall mean (↑ above, ↓ below).*

Metric Level	**Sessions**	**Precision Score**	**Recall Score**
Session-Level	↑ 1	**0.826**	0.775
	↓ 2	0.744	**0.784**
	↓ 3	0.756	0.750
	↓ 4	0.777	0.742
Test Dataset	Overall Mean	**0.776**	**0.763**
	Overall F1 Score	**0.769**	

The test dataset was split into five batches because it has 501 episodes of 4 sessions each, whereas the validation dataset has 500 episodes; the fifth batch contained a single episode (4 sessions). The scores are almost the same across the two datasets, with the highest precision again in session 1. On the test dataset the mean precision is slightly higher than on the validation dataset, and here precision slightly exceeds recall - the reverse of the validation dataset, where recall was higher than precision. Performance stays stable on the test dataset, which is consistent with the SVM model having learned the markers for locating persona facts (related to the Given-New distinction) rather than memorizing the training vocabulary. Taken together, the scores from both datasets suggest that the generated summaries preserve and recognize the utterances carrying the 'gold' persona facts, which matter for continuity across sessions.

4.6.2 BERTScore

Alongside the LLM judge, we also used BERTScore (Zhang et al., 2020) (model: deberta-xlarge-mnli), computing it for each of the four experiments. BERTScore com-

plements the LLM judge: it measures the similarity between the predicted summary and the 'gold' one, again comparing each session summary with its 'gold' counterpart, to see how semantically close the generated summaries are to the gold ones. The results for the validation dataset are in table 4.10.

Table 4.10: Session-level BERTScore results on the validation set. *Arrows compare each session's F1 with the overall mean (↑ above, ↓ below).*

Metric Level	Sessions	BERT Recall	BERT Precision	BERT F1-Score
Session-Level	↑ 1	**0.712**	**0.622**	**0.663**
	↓ 2	0.673	0.557	0.608
	↓ 3	0.665	0.544	0.598
	↓ 4	0.666	0.553	0.603
Validation Dataset Summary	Overall Mean	**0.679**	**0.569**	**0.618**

We repeated the process on the test dataset and the results are in are in table 4.11.

Table 4.11: Session-level BERTScore results on the test set. *Arrows compare each session's F1 with the overall mean (↑ above, ↓ below).*

Metric Level	Sessions	BERT Recall	BERT Precision	BERT F1-Score
Session-Level	↑ 1	**0.716**	**0.624**	**0.666**
	↓ 2	0.669	0.553	0.604
	↓ 3	0.659	0.545	0.596
	↓ 4	0.658	0.545	0.595
Test Dataset Summary	Overall Mean	**0.676**	**0.566**	**0.615**

The scores are very similar across the validation and test datasets. Like the LLM judge, BERTScore is consistent between the two datasets, which suggests that the summaries produced by the SVM pipeline preserve the semantic content.

Because this metric computes word-level similarity between the two summaries, it may not be the most appropriate choice for this dataset: BERTScore only looks at similarity between words, without understanding context or facts, whereas the LLM judge in addition acts as a fact-checker. The LLM judge checks whether a specific persona fact appears in the predicted summary in addition to similarity, while BERTScore only signals similarity and does not report fact coverage.

The LLM judge achieves higher scores than BERTScore. This is likely because the BERTScore measures semantic similarity rather than checking persona facts, and all words play an equal role and it does not consider compositionality: differences in wording and paraphrasing lower its scores even when both summaries convey the same facts. Similarly, because the summaries extracted by the SVM model contain the exact relevant utterances, they also include greetings, extra words and other material that is not a persona fact, and BERTScore penalizes this noise. For a task involving conversational memory, the LLM judge is more appropriate, because it ignores wording and recognizes the persona facts even when they are paraphrased.

4.7 Feature Ablation - Validation Dataset

After training the SVM model and running the experiments, the next step is feature ablation. The purpose is to measure the contribution of each feature by deconstructing the system to get a clearer picture of the contribution of each type of feature. The

study proceeds in four steps where the first step uses all of the features to test the performance of the model. The second step uses each feature individually, to understand the impact of each individual feature. The third step removes one feature at a time. This shows which combination works best and how the performance changes when one of the features is missing. The fourth step groups the features into two separate subsets according to a single principle: the highest precision or the highest recall for the 'relevant' class. We treat the 'not relevant' class as background noise, since the thesis is concerned with finding the linguistic cues that signal relevance in human-human dialogue, and we consider that recall of relevance is more important than recall of irrelevant. This is why we focus on the 'relevant' class for the fourth step of the study.

For the ablation study we report up to three decimal places, because in some cases the differences are very small and only become noticeable at that precision. The values for the four experiments are also given up to three decimals. The results for the four experiments are shown in the table 4.12.

Table 4.12: Feature ablation on the validation set: macro-averaged metrics by experiment. *Arrows: each experiment's macro F1-score vs. experiment 3 (↑ higher, ↓ lower, → similar).*

Experiment	Accuracy	Precision (Macro)	Recall (Macro)	F1 (Macro)	Precision (Rel.)	Recall (Rel.)	F1 (Rel.)
↓ Experiment 1	0.762	0.741	0.747	0.744	0.826	0.800	0.813
↓ Experiment 2	0.772	**0.764**	0.719	0.731	0.780	**0.900**	**0.836**
→ Experiment 3	0.767	0.746	0.753	0.749	0.832	0.800	0.816
↑ Experiment 4	**0.776**	0.756	**0.766**	**0.760**	**0.845**	0.799	0.821

We compare the ablation results against experiment 3, because experiment 4 also includes the 'text utterances'. The aim is to isolate the impact of the core linguistic features, which excludes the 'text utterances'.

In the second step, the model is restricted to one feature at a time, which lets us evaluate each feature in isolation and assess its raw capacity to predict the 'relevant' utterances. The next two tables give an overview of each feature used individually, from the macro-average perspective and from the 'relevant'-class perspective. The table 4.13 reports the macro averages.

Table 4.13: Single-feature ablation on the validation set: macro-averaged metrics. *Note: the validation dataset contains 25,459 utterances in total, of which 16,432 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each feature's macro F1-score alone vs. experiment 3 (↑ higher, ↓ lower, → similar).*

Isolated Feature	Predicted Relevant	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
→ Full set (Experiment 3)	15,796	0.767	0.746	0.753	0.749
↓ Position	25,459	0.645	0.323	0.500	0.392
↓ Agency	15,208	**0.751**	**0.732**	**0.744**	**0.736**
↓ Possession	25,338	0.643	0.463	0.499	0.395
↓ Interjection	24,099	0.648	0.590	0.520	0.458
↓ Discourse Marker	25,454	0.645	0.523	0.500	0.392
↓ Emotion	21,727	0.637	0.569	0.538	0.517
↓ Negation	25,459	0.645	0.323	0.500	0.392
↓ Temporal	25,459	0.645	0.323	0.500	0.392

The overview table 4.14 for the 'relevant' class follows.

Table 4.14: Single-feature ablation on the validation set: relevant-class metrics. *Note: the validation dataset contains 25,459 utterances in total, of which 16,432 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each feature's 'relevant'-class F1-score alone vs. experiment 3 (↑ higher, ↓ lower, → similar).*

Isolated Feature	**Precision (Rel.)**	**Recall (Rel.)**	**F1-Score (Rel.)**
→ Full set (Experiment 3)	0.832	0.800	0.816
↓ Position	0.645	**1.000**	0.785
↓ Agency	**0.832**	0.770	**0.800**
↓ Possession	0.645	0.995	0.783
↓ Interjection	0.655	0.961	0.779
↓ Discourse Marker	0.645	**1.000**	0.784
↓ Emotion	0.666	0.880	0.758
↓ Negation	0.645	**1.000**	0.785
↓ Temporal	0.645	**1.000**	0.785

Here, 'agency phrases' stands out as the strongest single feature and the one with the most impact on the model. Used on its own, its macro averages for precision, recall and F1-score are 0.732, 0.744 and 0.736; for the 'relevant' class, the precision, recall and F1-score are 0.832, 0.770 and 0.800. The step also shows that features such as 'emotion polarity', 'interjection', 'possessive phrases', 'discourse markers', 'position', 'negation' and 'temporal' have almost no predictive power on their own. Each of them labels nearly every utterance in the validation dataset as relevant - between 21,727 and 25,459 of the 25,459 total - while the gold set contains 16,432 relevant utterances. A likely reason is that these features act as modifiers rather than anchors, so they cannot be informative by themselves. This does not mean they are useless, only that they cannot stand alone, and it suggests that relevance in dialogue cannot be captured by single linguistic cues; it requires a combination of several interacting cues.

In short, 'agency phrases' is the most informative standalone feature, while the others contribute little on their own.

The next step, 'minus one feature', looks at the effect of removing a single feature from the full set. The features are 'agency phrases', 'possessive phrases', 'interjection', 'discourse markers', 'negation', 'emotion' (emotion polarity and emotion words), 'temporal' (temporal state and trigger word) and 'position' (which combines 'session stage' and 'turn stage'). The overview of the omitted features for the macro averages is in table 4.15.

Table 4.15: Leave-one-feature-out ablation on the validation set: macro-averaged metrics. *Note: the validation dataset contains 25,459 utterances in total, of which 16,432 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: change in macro F1-score when each feature is removed, vs. experiment 3 (↑ improvement, ↓ decline, → negligible).*

Omitted Feature	Predicted Relevant	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
→ Full set (experiment 3)	15,796	0.767	0.746	0.753	0.749
→ Position	15,808	0.766	0.745	0.752	0.748
↓ Agency	17.627	0.669	0.632	0.623	0.626
↓ Possession	15,582	0.740	0.719	0.727	0.722
→ Interjection	15,785	0.767	0.746	0.753	0.749
→ Discourse Marker	15,808	0.767	0.746	0.753	0.749
↑ Emotion	15,831	**0.775**	**0.755**	**0.762**	**0.758**
→ Negation	15,840	0.768	0.748	0.754	0.751
→ Temporal	15,815	0.768	0.747	0.754	0.750

The overview table 4.16 for the 'relevant' class follows.

Table 4.16: Leave-one-feature-out ablation on the validation set: relevant-class metrics. *Note: the validation dataset contains 25,459 utterances in total, of which 16,432 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: change in the 'relevant'-class F1-score when each feature is removed, vs. experiment 3 (↑ improvement, ↓ decline, → negligible).*

Omitted Feature	Precision (Rel.)	Recall (Rel.)	F1-Score (Rel.)
→ Full set (experiment 3)	0.832	0.800	0.816
→ Position	0.831	0.799	0.815
↓ Agency	0.727	0.780	0.752
↓ Possession	0.815	0.773	0.793
→ Interjection	0.832	0.800	0.816
→ Discourse Marker	0.832	0.801	0.816
↑ Emotion	**0.838**	**0.808**	**0.823**
→ Negation	0.833	0.803	0.817
→ Temporal	0.833	0.802	0.817

Removing 'agency phrases' causes the scores to drop sharply, confirming the central role of this feature, which can be called the 'anchor' of the system. Without it, the model over-predicts the relevant class, labeling 17,627 utterances as relevant when the gold set has 16,432. The macro averages for precision, recall and F1-score fall to 0.632, 0.623 and 0.626. The 'relevant' class scores also differ from those of experiment 3, with precision, recall and F1-score of 0.727, 0.780 and 0.752. The model loses its selective behavior, and its performance degrades. The second most important feature appears to be 'possessive phrases'. When it is dropped, the macro averages for precision, recall and F1-score fall to 0.719, 0.727 and 0.722, and the 'relevant' class precision, recall and F1-score to 0.815, 0.773 and 0.793. These results indicate that the two features capture core information about the persona and are useful for signaling relevance in multi-turn dialogue. They also suggest that relevance in conversations spanning multiple sessions centers on information about actions, intentions, references to oneself and ownership.

The remaining features behave neutrally. Removing 'position', 'interjection', 'discourse markers', 'negation' or 'temporal' leaves the scores almost unchanged, so their individual effect on the model is small. The ablation also gives some insight into the 'emotion polarity' feature, which appears to be redundant. Leaving it out improves the model slightly: the macro averages for precision, recall and F1-score rise to 0.755, 0.762

and 0.758, compared with 0.746, 0.753 and 0.749 for experiment 3. The 'relevant'-class values stay close to those of experiment 3. This could point to the emotion feature somehow counteracting and reducing the effectiveness of other feature combinations.

The final step creates two feature subsets, labeled 'high precision' and 'high recall', based on the 'relevant' class. We use these values because the thesis is concerned with what is relevant in human-human dialogue and can be reused in future interactions. The two subsets are:

- High precision: 'agency phrases', 'emotion' and 'interjection' - the features whose 'relevant'-class precision is above the baseline of 0.645.
- High recall: 'position', 'possessive phrases', 'discourse markers', 'negation' and 'temporal' - the features whose 'relevant'-class recall falls between 0.995 and 1.000.

The macro-average results are in table 4.17.

Table 4.17: Feature-subset ablation on the validation set: macro-averaged metrics. *The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each subset's macro F1-score vs. experiment 3 (↑ higher, ↓ lower, → similar).*

Feature Subset	Total Utterances	Gold Relevant	Predicted Relevant	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
→ Full set (Experiment 3)	25,459	16,432	15,796	0.767	0.746	0.753	0.749
↓ High Precision	25,459	16,432	15,523	**0.736**	**0.715**	**0.723**	**0.718**
↓ High Recall	25,459	16,432	20,304	0.650	0.599	0.570	0.565

For the 'relevant' class the results are in table 4.18.

Table 4.18: Feature-subset ablation on the validation set: relevant-class metrics. *Note: the validation dataset contains 25,459 utterances in total, of which 16,432 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each subset's 'relevant'-class F1-score vs. experiment 3 (↑ higher, ↓ lower, → similar).*

Feature Subset	Precision (Rel.)	Recall (Rel.)	F1-Score (Rel.)
→ Full set (Experiment 3)	0.832	0.800	0.816
↓ High Precision	**0.813**	0.768	**0.790**
↓ High Recall	0.685	**0.847**	0.758

The high-precision subset under-predicts the relevant class, labeling 15,523 utterances as relevant out of 16,432 gold, with a macro F1-score of 0.718 and a 'relevant' class F1-score of 0.790. This suggests that the classifier keeps a strict decision boundary and is fairly robust to the noise in the dataset. The high-recall subset over-predicts, labeling 20,304 utterances as relevant. Its macro F1-score is 0.565 and its 'relevant' class F1-score is 0.758. This confirms that some features are interdependent and cannot stand on their own: the absence of 'agency phrases', the anchor of the system, has a large effect, and the decision boundary collapses into over-prediction.

4.8 Feature Ablation - Test Dataset

We will now repeat some of the high-value steps from the feature ablation study on the validation dataset to check whether the same pattern holds true for the test set. On the validation dataset, 'agency phrases' had the largest impact on performance, followed

by 'possessive phrases'. We call the remaining features 'neutral', because they cannot stand on their own and removing them from the set has little or no noticeable effect. The exception was 'emotion polarity', whose removal improved the model. With this in mind, we ran the feature ablation on the test dataset to see whether these patterns are consistent.

On the test dataset, the patterns found in the validation study still hold true. The table 4.19 gives the full overview of the individual features.

Table 4.19: Single-feature ablation on the test set: macro-averaged metrics. *Note: the test dataset contains 24,375 utterances in total, of which 16,397 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each feature's macro F1-score alone vs. experiment 3 (↑ higher, ↓ lower, → similar).*

Isolated Feature	Predicted Relevant	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
→ Full set (Experiment 3)	15,104	0.758	0.730	0.746	0.735
↓ Position	24,375	0.673	0.336	0.500	0.402
↓ Agency	14,620	**0.746**	**0.721**	**0.741**	**0.726**
↓ Possession	24,266	0.670	0.446	0.499	0.404
↓ Interjection	23,054	0.674	0.599	0.523	0.473
↓ Discourse Marker	24,368	0.673	0.479	0.500	0.402
↓ Emotion	20,741	0.652	0.561	0.535	0.521
↓ Negation	24,375	0.673	0.336	0.500	0.402
↓ Temporal	24,375	0.673	0.336	0.500	0.402

For the 'relevant' class the individual features are in table 4.20.

Table 4.20: Single-feature ablation on the test set: relevant-class metrics. *Note: the test dataset contains 24,375 utterances in total, of which 16,397 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: each feature's 'relevant'-class F1-score alone vs. experiment 3 (↑ higher, ↓ lower, → similar).*

Isolated Feature	Precision (Rel.)	Recall (Rel.)	F1-Score (Rel.)
→ Full set (Experiment 3)	0.847	0.781	0.813
↓ Position	0.673	**1.000**	**0.804**
↓ Agency	**0.849**	0.757	0.801
↓ Possession	0.672	0.995	0.802
↓ Interjection	0.683	0.961	0.799
↓ Discourse Marker	0.673	**1.000**	**0.804**
↓ Emotion	0.691	0.874	0.772
↓ Negation	0.673	**1.000**	**0.804**
↓ Temporal	0.673	**1.000**	**0.804**

Again, 'agency phrases' is the only feature that does not over-predict, while the others label more utterances as relevant than they should. Every feature other than 'agency phrases' over predicts utterances as relevant. This again shows that these features cannot stand on their own and need to be combined to curb over-prediction.

The values are similar to those from the validation dataset study, with only minor differences. For the features we consider neutral, the effect of removing them from the full set is so small that it cannot be read as a negative impact; they are not directly harmful and the differences are not substantial.

On the test dataset, we focus on whether 'emotion polarity' still harms the model and whether removing 'agency phrases' or 'possessive phrases' degrades it. The values

show the same pattern as on the validation dataset. We again compared against the same experiment 3 model used for the validation set in table 4.21, and this time we also noticed a difference for the 'interjection' feature.

Table 4.21: Leave-one-feature-out ablation on the test set: macro-averaged metrics. *Note: the test dataset contains 24,375 utterances in total, of which 16,397 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: change in macro F1-score when each feature is removed, vs. experiment 3 (↑ improvement, ↓ decline, → negligible).*

Omitted Feature	Predicted Relevant	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
→ Full set (experiment 3)	15,104	0.758	0.730	0.746	0.735
↓ Agency	16,737	0.671	0.623	0.621	0.622
↓ Possession	14,920	0.732	0.703	0.719	0.708
↑ Emotion	15,154	**0.767**	**0.739**	**0.755**	**0.745**
↑ Interjection	15,101	0.763	0.735	0.752	0.741

The overview table 4.22 for the 'relevant' class follows.

Table 4.22: Leave-one-feature-out ablation on the test set: relevant-class metrics. *Note: the test dataset contains 24,375 utterances in total, of which 16,397 are gold relevant utterances. The top row is experiment 3 (the full feature set), shown for comparison. Arrows: change in the 'relevant'-class F1-score when each feature is removed, vs. experiment 3 (↑ improvement, ↓ decline, → negligible).*

Omitted Feature	Precision (Rel.)	Recall (Rel.)	F1-Score (Rel.)
→ Full set (experiment 3)	0.847	0.781	0.813
↓ Agency	0.750	0.766	0.758
↓ Possession	0.830	0.756	0.791
↑ Emotion	**0.853**	**0.789**	**0.820**
↑ Interjection	0.852	0.784	0.817

When 'emotion polarity' and 'interjection' are removed, the model improves, which suggests that these features have a negative effect and may introduce noise. Unlike on the validation dataset, where 'interjection' was neutral rather than negative, here it has an effect, which may point to the inconsistency of this feature. One reason may be the data: in a natural setting these filler words are noisy and inconsistent, and they do not carry stable information that contributes to long-term memory and persona. By contrast, 'agency phrases' and 'possessive phrases' are consistent across the dataset and on unseen material. Removing either feature lowers performance, most sharply for 'agency phrases', which makes them the most important components of the study. When 'agency phrases' is removed, the model also begins to over-predict, which suggests that this feature acts both as a constraint and as an indicator for the model.

We conclude that 'agency phrases' and 'possessive phrases' are the most impactful, informative and useful features, which is consistent with the tendency of people to talk about themselves. In dialogue, these two features signal and encode self-related information such as likes, dislikes, actions, relationships, thoughts, ownership and dependencies.

4.9 Error Analysis

In the feature ablation study on test dataset, we found that 'emotion polarity' has a negative impact. This finding is consistent across both datasets, validation and test. When the feature is removed from the features, the performance of the model improves compared to the model with all features (structured cues with cue text n-grams). The goal of this error analysis is to examine some of these cases and understand what kind of errors the emotional information introduces. We first compared the predictions between the model the model with all features (structured cues with cue text n-grams) and the model with all features except the 'emotion polarity' features. Then, we selected the samples in the test dataset where there were disagreements between the prediction of these two models (different predictions). We randomly chose a subset of 50 sample utterances from all the selected disagreements for the purpose of error analysis.

The utterances were evaluated manually to investigate what kind of errors occur in association with the 'emotion polarity' feature. We chose not to include the 'interjection' feature in this error analysis: its effect was not consistent across the datasets, as it degraded performance only on the test dataset, whereas the behavior of 'emotion polarity' was consistent.

As explained in earlier section, the emotion labels originate from the VADER annotation, and from (Table B.1 Sampled Utterances in the Appendix B) it is clear that 'strong negative' does not appear in any of the samples, while 'strong positive' appears in 24 of the cases and 'neutral' in 26. In 17 of the 50 cases, the model with all features (structured cues with cue text n-grams) predicted 'True' (column 6) while the model with all features except 'emotion polarity' (column 7) predicted 'False', which matches the 'gold' annotation. These are false positives: the model with all features predicted 'relevant' while the 'gold' was 'not relevant', and once the 'emotion polarity' feature was removed, the model predicted correctly. In 33 of the 50 cases, the model with all features (structured cues with cue text n-grams) predicted 'False' while the model with all features except 'emotion polarity' predicted 'True', matching the 'gold'. These are false negatives: the model with all features predicted 'not relevant' while the 'gold' was 'relevant', and once the feature was removed, the model matched the 'gold' as 'relevant'. This suggests, based on our selected subset, that the model with 'emotion polarity' predicts 'not relevant' more often than it should. The pattern can be described as the 'emotion polarity' feature tending to suppress 'relevant' predictions more often than it helps identify utterances that contain factual information. As we will see later, this pattern is also consistent with negative coefficients for all emotion polarity categories in the model with all features (structured cues with cue text n-grams), signaling that this feature nudges the model to predict the negative class (not relevant). This is one possible explanation for the slight improvement in performance when this feature is removed.

We examined the false positives and false negatives in all of the 1,331 utterances with disagreement on relevance prediction between the two aforementioned models, and it showed that this same pattern holds true across all the disagreements. False positives account for 466 of the 1,331 examples (35.01%), while false negatives account for 865 (64.99%). These numbers represent the overall behavior of the model. Again, this indicates that, when 'emotion polarity' is involved, the model tends to miss 'relevant' utterances rather than over-predict them, which is consistent with the small sample selected for the error analysis.

In the small sample, we do not see a clear pattern related to the type of emotion, but we can observe that 'strong positive' tends to occur when the utterances are more expressive, while 'neutral' occurs when they are more informational or descriptive.

For example, in 'test_445', the utterance 'They're doing well. They're telling me about all this pop music that they like but I don't know any of the bands. Do you know any pop bands' has the 'gold' 'I don't know any of the new pop bands that they like. I have grandkids.' The model assigned this utterance a 'neutral' label. We also observe that the 'gold' is not entirely correct, as it contains extra information that is not found in the utterance. The utterance was classified as 'neutral' because it states factual information without being expressive, so the model overlooked it (a false negative) and classified it as 'not relevant'. Once 'emotion polarity' is removed, the model is able to recognize the 'relevant' information.

In another example, 'test_288', the utterance 'Really?! I didn't know that! I sure hope you will be able to make it, especially now that you have offered to help with the meat ;)' has no persona fact in the 'gold' annotation and is labeled with the emotion 'strong positive'. The words in the sentence ('hope', 'help') misled the SVM model into 'deciding' that the utterance contained important factual persona information, so it predicted 'relevant' (a false positive) when there was no persona fact. Once 'emotion polarity' was removed, the model predicted 'not relevant' and matched the 'gold'.

The next step in understanding why 'emotion polarity' led the SVM model away from predicting 'relevant' utterances was to inspect the coefficients of this feature. This helps reveal the impact of the feature within the SVM model and provides insight into the learned feature weights. We found that all three categories of 'emotion polarity' have a negative coefficient in table 4.23.

Table 4.23: Emotion polarity coefficients and their impact on the SVM model.

Emotion polarity	**Value**
Strong positive	-0.13
Neutral	-0.07
Strong Negative	-0.05

Inspecting the coefficients, we see that the weight points to negative class (not relevant) for all three categories, with the 'strong positive' category having the strongest impact. This supports the ablation study and likely causes the recall for the 'relevant' class to drop when the emotion feature is present. There are various possible explanations: either the emotional language is not the best indicator of factual persona relevance in the multi-turn dataset, or there are annotation errors in the dataset, or emotion polarity feature was miscalculated.

On the other hand, for the full model with all features (structured features together with cue-text n-grams) in figure 4.2, the highest weights for the positive class ('relevant') and the negative class ('not relevant') go to certain cue-text n-grams. It may suggest that the model considers the specific semantic information in the linguistic cues more important for classification.

```
Top 5 positive features for relevance
hah glad 1.5979319201825906
heard bet 1.5596154441543906
unfortunate recently 1.434721900741457
am well glad 1.4314938632436103
aww sweet soon 1.4166863552674558

Top 5 negative features for relevance
yeah you know best -1.6958384648775962
ride nope -1.6667273874498951
do hey lol -1.6399591052437152
well well soon -1.6207234598808165
do am currently -1.594142742293652
```

Figure 4.2: The top five positive and negative cue-text n-gram features for the relevant class, with their SVM coefficients.

4.9.1 Emotion Polarity Feature Validation

Because VADER measures vocabulary and computes scores from a dictionary, we decided to manually add our own emotion label to the 50 previously selected examples for error analysis with VADER based emotion label. This is done mainly to capture human intent and the overall sentiment of each utterance without relying on a dictionary-based algorithm. We took this step because we suspected that some of the VADER labels are not fully correct and that some sentiments may be confused with one another. Another reason is that the feature ablation suggested that 'emotion polarity' is redundant, so manual labeling might give us better insight into why. Yet another reason 'emotion polarity' did not improve the SVM model might be that the three categories are too broad, forcing VADER to categorize words that do not fit neatly. The three categories are: neutral, strong positive, and strong negative.

For the human assessment we used the following criteria:

- Neutral: for utterances that do not have a clear emotional tone; they are mostly informative or express general opinions about a situation and lack emotional intensity.
- Strong positive: for utterances that express enthusiasm, joy, or positive life experiences.
- Strong negative: for utterances that describe a negative life event or a sad situation, or that express clear dislike, frustration, or sadness.

The label was assigned based on the context and the emotional tone inferred from the text. In addition, as a further step in order to get another viewpoint, we provided the 50 samples to an LLM and asked it to generate an emotion label (Prompt Emotion Classification in the Appendix B). The comparisons were human-VADER, human-LLM, and VADER-LLM (the last three columns of the B.1 Sampled Utterances in the Appendix B contain the VADER, human, and LLM labels).

As before, we used Gemini 3.1 Pro Preview for this experiment, accessed through Google AI Studio. We set the temperature to 0.0, the thinking level to high, and top-p to 0.1, so that the evaluation would be deterministic. The following table 4.24 contains the disagreements and agreements among the three sets of annotations.

Table 4.24: Agreement and disagreement among the VADER, human, and LLM emotion labels on the 50 samples.

Comparison group	All disagree	Partial agree (2 out of 3)	All agree
VADER - Human	16	-	34
VADER - LLM	15	-	35
LLM - Human	11	-	39
LLM - Human - VADER	2	18	30

The first thing we notice is that all three annotations match in 30 of the 50 cases. This can indicate that, when emotions are evident, both VADER and the LLM are able to identify them. The LLM and the human agree on the same label more often (39 cases) than VADER and the human (34 cases) or VADER and the LLM (35 cases). VADER matched the human 68% of the time, while the LLM matched the human 78% of the time. This suggests that the LLM is better than VADER at understanding the context captured in the samples. Even so, the LLM and the human still disagree in 11 cases.

Another interesting aspect is to look at the two samples on which all three annotations disagreed. The utterances are:

- I hope she is nicer to her kids. Luckily she doesn't have a daughter to call fat. How old are your nephews? They must be so cute! (VADER - strong positive, human - neutral, LLM - strong negative).
- I know what you mean. I never have any alone time. What will we do when we have freedom? (VADER - strong positive, human - neutral, LLM - strong negative).

We see the same pattern in both cases: VADER assigned strong positive, the human assigned neutral, and the LLM assigned strong negative. In these two cases VADER appears to be misled by words with positive connotations, such as 'hope', 'cute', and 'freedom', whereas the human label may indicate that the emotions are not strong enough to be labeled strong positive or strong negative. From the human point of view, both samples show that complaining about general life issues does not always imply a strong emotion and can be read as 'gossip'. A human may consider them chit-chat without dramatizing their meaning. The LLM, in contrast, assigned strong negative to both samples, which may indicate an over-analysis of the words and the context. In the first case, we can assume the LLM assigned strong negative because of the safety filters built into its mechanism.

Another point to note, in the 50 samples, VADER never assigned strong negative, the human assigned it 4 times, and the LLM assigned it in 6 cases; the human and the LLM both assigned strong negative in 3 cases. In those 3 cases, VADER assigned neutral twice and strong positive once. This may indicate that VADER misses the nuance of some samples, while the LLM over-assigns strong negative because of an over-sensitivity to safety issues. It may also indicate that VADER is less sensitive to negative nuance in the utterances, especially when the negativity is implied, indirect, or pragmatic rather than stated directly. This analysis has its own limitation though, since it is focused only on 50 samples and might not reflect the whole dataset. The trend of lower proportion for strong negative polarity compared to others was also reported in the data exploration section earlier for both test (1.95% for strong negative

vs 35.07% for strong positive and 62.98% for neutral) and training dataset (2.05% for strong negative vs 36.14% for strong positive and 61.81% for neutral). This suggests that strong negative is underrepresented in the corpus.

Overall, we can argue that VADER does not always capture the context of the utterances, which might be one reason the 'emotion polarity' feature quality is not sufficient and may introduce noise into the SVM's relevance predictions.

4.9.2 Human Validation

Next, we performed human validation by manually classifying the relevance of each of the previously selected 50 utterances for error analysis in test dataset (columns 3, 4, and 5 of Table B.1 Sampled Utterances in the Appendix B, with the text in column 2). This step was introduced to critically check the integrity of the 'gold' labels and to better understand the semantic ambiguity that can exist in the test dataset. The procedure was to read each sample, decide whether it is relevant or not relevant, and then examine the disagreements and agreements among the human, the 'gold', and the combined SVM model (the model that includes the full feature set and text utterances). The results are as follows in table 4.25.

Table 4.25: Agreement and disagreement between human validation, the 'gold', and the full SVM model on the 50 sampled utterances.

Comparison group	**All disagree**	**Partial agree (2 out of 3)**	**All agree**
Human - Gold	7	-	43
Human - SVM	29	-	21
SVM - Gold	30	-	20
Human - Gold - SVM	-	33	17

The smallest disagreement is between human validation and the 'gold', with 7 disagreements and 43 agreements on whether a sample is relevant for long-term conversation. These 7 human-human disagreements can indicate subjectivity in what counts as relevant. This brings us back to relevance theory, in which relevance is treated as a 'fuzzy' concept that can vary from person to person and from situation to situation. The largest disagreement is between the combined SVM model and the 'gold' (30 cases), followed by human-SVM (29 cases). Only in 17 cases do all three sources agree on the same label, 'relevant' or 'not relevant'. This suggests that relevance classification can be subjective in some cases and that the SVM model misclassified in some cases.

Next, we collected all 7 examples in which human validation and the 'gold' disagree. They are shown in the following table 4.26.

Table 4.26: The seven utterances where human validation and the 'gold' disagree; the samples marked from 2 to 7 have summaries that draw on a wider context than the utterance itself.

Text	**Human validation**	**Gold**
Yes, there is no way I could have left him, I am happy to give him a home. Have you ever rescued an animal?	Relevant	Not Relevant
Yeah...if you don't like the movies marvel makes then you have been pretty much out of luck the past 12 years	Not Relevant	Relevant
I'm so glad to hear that! Is he your first dog or have you had other pets before?	Not Relevant	Relevant

Did you find a good car on one of those auto websites I sent to you?	Not Relevant	Relevant
They'll teach you everything there much better than I could haha, but just know that it's not as bad as it looks! Just remember to enjoy yourself and that an accident is extremely unlikely.	Not Relevant	Relevant
I'm so glad that you liked it! I watched it only once but I've thought about watching it again. Well if you tried one of mine, maybe I'll watch one of yours.	Not Relevant	Relevant
I think so- it's kind of like the Pacific Coast highway of Scotland right?	Not Relevant	Relevant

The samples from 2 to 7 are those whose summaries contain information that does not align perfectly with the utterance, or that include information from earlier utterances (this is discussed further later in the thesis).

The first thing we notice is that, in the human evaluation, 6 of the 7 samples are marked as not relevant while the 'gold' from the test dataset marks them as relevant, which may indicate false positives generated by the crowd workers. The single remaining sample is marked as relevant in the human validation but as not relevant by the crowd workers. For the 6 samples marked as relevant by the 'gold', we will see later that the extracted persona and the relevance are not fully supported by the utterance, and that the crowd workers used a wider conversational context window. Only one disagreement is a false negative, where the crowd workers overlooked explicit long-term information: in the sample 'Yes, there is no way I could have left him, I am happy to give him a home. Have you ever rescued an animal?', the information that an animal was rescued is an important and memorable persona fact.

4.9.3 Annotation Errors

During the human validation of the 50 samples and while looking at the extracted persona, we noticed some mismatches between the text and the persona information (Table 4.27). The personas do not always align with the utterance, which may indicate that in some cases the annotators used a larger context window; this can mislead the SVM model, which predicts 'relevant' and 'not relevant' at the utterance level. It may also explain why some of the errors made by the SVM model and even by the LLM judge are due to the annotations. We found 18 of the 50 cases (36%) that contain this issue, which may reflect a pattern present in the whole dataset and may be one reason the F1-score is penalized. The annotators often append contextual information they learned earlier to the current utterance. Because the SVM model is trained at the utterance level, it is bound to the immediate text string: it cannot recover information such as 'skydiving' if it is not in the text, and it can be affected by noisy data and annotation.

Two examples from the table help illustrate how some extracted personas are not fully related to the samples. The first is: 'Oh, we're not living here, we just wanted to make a trip out and see it. The parking is a bit difficult with my Subaru, however.', for which the extracted persona is 'I just bought a new Subaru.' The summary is not fully supported by the sample; a better version might be 'It's difficult to park with my Subaru.' This indicates that the annotators used a broad context window, since the sample does not explicitly state whether the Subaru is new. Some of the annotations may therefore come from a broader earlier context.

The second example is: 'Haha no clue, but I am pretty excited about it. It will definitely help me to feel closer to my neighbors.', for which the extracted persona is 'I am excited about our new Spanish tradition.' Again, the summary is not fully supported by the sample; a better version might be 'I want to feel closer to my neighbors.' The sample makes no mention of Spanish traditions, so this is likely information that was mentioned at some other point in the conversation.

We can argue that, when summarizing the utterances, the crowd workers sometimes merged details from previous utterances into the current one. This may indicate that some of the model's errors are due to the alignment between the annotations and the utterances, which points to a limitation of utterance-level analysis. This is a disadvantage for the SVM model, and also when the LLM is used as a judge: if an isolated utterance is fed into the model, it is scored against a 'gold' that contains external context to which the model has no access.

Table 4.27: Sampled utterances whose extracted persona includes information not stated in the utterance (18 of the 50 cases).

Initial data id	**Text**	**Extracted persona**
15135 test_445	They're doing well. They're telling me about all this pop music that they like but I don't know any of the bands. Do you know any pop bands?	I don't know any of the new pop bands that they like. I have grandkids.
22374 test_698	Well It keeps me bust because I clean the house too. The kids are fun but one of them is a little crazy. He bit me yesterday!	I work as a babysitter. One of the kids I babysit bit me. I also clean the house.
20545 test_364	Haha no clue, but I am pretty excited about it. It will definitely help me to feel closer to my neighbors.	I am excited about our new Spanish tradition.
16134 test_644	Yeah...if you don't like the movies marvel makes then you have been pretty much out of luck the past 12 years	I think the Marvel franchise has over saturated the market.
23826 test_908	I'm so glad to hear that! Is he your first dog or have you had other pets before?	I'm glad the dog is fine.
10300 test_657	I think hiking is more my style but I will look into the donkey's. Did you travel with anyone when you went?	I think I would rather hike that ride a donkey.
20966 test_432	Did you find a good car on one of those auto websites I sent to you?	I know a good amount about cars.
10291 test_655	They'll teach you everything there much better than I could haha, but just know that it's not as bad as it looks! Just remember to enjoy yourself and that an accident is extremely unlikely.	I cannot teach skydiving.
22074 test_645	Oh, we're not living here, we just wanted to make a trip out and see it. The parking is a bit difficult with my Subaru, however.	I just bought a new Subaru.
22549 test_723	Is there anything I can do to help? I've been in that position before and I can only imagine that the down swing is very rough financially.	I have tried to work freelance in the past.
24140 test_957	I'm so glad that you liked it! I watched it only once but I've thought about watching it again. Well if you tried one of mine, maybe I'll watch one of yours.	I like the Jimmy Car special.
13217 test_103	Yes! I actually did the west highland way hike. It was so beautiful! That trip made me want to adventure even more	I like to adventure. I have been hiking in Scotland.

4683 test_739	I love it here. Would love to see greece someday.	I love Florida. I would like to visit Greece.
10161 test_637	I know what you mean. I never have any alone time. What will we do when we have freedom?	I want my own apartment. I want alone time.
14352 test_310	I find them interesting in an almost historical sense - why people follow certain religious ideas, for example	I like the historical context behind the bible.
17017 test_784	I find it so calming. One of the reasons I love San Diego. The Gaslight District is really fun, too. How long is your drive going to be?	I have visited San Diego before.
17995 test_924	Maybe working for international aid or something like that. I'm not sure yet, I should start looking!	I'd like to work for an international non-profit.
13219 test_103	I think so- it's kind of like the Pacific Coast highway of Scotland right?	I know the Pacific Coast Highway in the USA.

4.10 Predicting Relevance per Utterance with LLM

Another experiment we conducted was to use an LLM to predict relevance at the utterance level on the test dataset, as a relevance-validation task. We wanted to check whether the utterances that our SVM full model considered relevant were also identified as relevant by the LLM, and to compare both with the 'gold' from the dataset (the 'extracted persona' created by the crowd workers). Because the crowd workers used a larger context for some utterances, as we noticed in the error analysis, this experiment can also show whether the annotators were subjective in deciding what is relevant in a conversation. For this task, we used Claude rather than Gemini 3.1 Pro Preview as Google AI Studio didn't work reliably with larger context and the task was not feasible, however, with Claude, we were not able to set the temperature or top-p. We instructed Claude to decide, for each utterance, whether it contains persona facts worth keeping for a long-term conversation (Prompt Utterance Relevance Prediction in the Appendix B). Claude was then instructed to predict TRUE for an utterance that contains relevant information and FALSE otherwise. The output was saved to a CSV file. From the predictions we created three scenarios: in scenario 1, the crowd workers who annotated the whole dataset are the 'gold' and the LLM is the prediction; in scenario 2, the LLM is the 'gold' and the SVM full model is the prediction; in scenario 3, the LLM is the 'gold' and the crowd workers are the prediction. The results for the three scenarios are as follows in table 4.28.

Table 4.28: Per-class and macro-averaged precision, recall, F1-score, and support for the three relevance-prediction scenarios.

Scenario	Class / avg	Accuracy	Precision	Recall	F1-score	Support
Scenario 1: Crowdworker (gold) vs. LLM (prediction)	Macro avg	**0.81**	0.83	0.74	**0.76**	24,375
	relevant		0.81	**0.95**	**0.87**	16,397
	not relevant		0.85	0.53	**0.65**	7,978
Scenario 2: LLM (gold) vs. SVM full model (prediction)	Macro avg	0.78	0.71	0.81	0.73	24,375
	relevant		**0.96**	0.75	0.84	19,413
	not relevant		0.47	0.87	0.61	4,962
Scenario 3: LLM (gold) vs. Crowdworker (prediction)	Macro avg	**0.81**	0.74	0.83	**0.76**	24,375
	relevant		0.95	0.81	**0.87**	19,413
	not relevant		0.53	0.85	**0.65**	4,962

The first notable observation concerns the class distribution. The LLM considers 79.64% utterances relevant compared to the 67.27% relevant gold 'extracted persona'. This is 18% higher for LLM prediction, and may indicate that the LLM used a broader

definition of relevance and tends to classify more utterances as relevant. It might also reflect a difference between the LLM and humans in how they perceive relevance in a long-term conversation. There may also be an issue with how the annotation was created manually: this is a very repetitive task, and the crowd workers may have skipped some utterances, or used a larger context and added a summary to one utterance that includes extra information from earlier turns.

In scenario 1, the LLM achieved a recall of 0.95 for the relevant class, but we notice a mirror effect in scenario 3, where the precision for the relevant class is 0.95. This happens because swapping the 'gold' and the predictions inverts the precision and recall formulas. We see the same effect for the not relevant class and for the macro-averaged recall and precision in scenarios 1 and 3. The high recall in scenario 1 for the relevant class indicates that the LLM almost never missed the persona facts identified by the crowd worker annotators. It can also indicate that the LLM additionally extracted information that the crowd workers considered noisy, as reflected in the low recall of 0.53 for the not relevant class. Scenarios 1 and 3 show that the LLM labels and the crowd worker labels are very close, with an F1-score of 0.76 in both cases. The main difference between the two scenarios may be that the crowd workers were more selective about relevance, as reflected in the higher precision than recall for the relevant class (0.95 versus 0.81). We can consider the crowd worker labels to be stricter than those of the LLM. The mirror effect can indicate the difference between human and machine behavior. Scenario 2 shows that the combined SVM model is close to the LLM labels, but the LLM is closer to the crowd worker labels. The model's performance on the relevant class is: precision 0.96, recall 0.75, and F1-score 0.84. This indicates that the SVM model is able to capture the relevant signal and to extract relevant persona information that can be used in long-term conversation. The low precision of 0.47 for the not relevant class may indicate that the SVM model struggles with false positives due to noisy or ambiguous linguistic cues (e.g. overlapping cues, annotation or context mismatch). For all three scenarios, the macro-averaged F1-scores are very similar: 0.76 for scenario 1, 0.73 for scenario 2, and 0.76 for scenario 3. This suggests that the three systems are roughly similar and that none is drastically better or worse than the others. We can argue that the relevance signal is stable across the scenarios and that they capture similar general patterns of relevance in long-term memory, with small differences in class balance.

Chapter 5

Discussion

The thesis aimed to find the linguistic cues and signals that can indicate relevance in human-human dialogue, using MSC:PersonaSummary, a subset of the multi-turn MSC dataset created by Facebook AI. The evaluation, feature ablation, and error analysis together provide insight into the behavior and complexity of this human-annotated dataset.

In the first part of the thesis, we focused on finding linguistic theories that could form a framework for the study: relevance theory, common ground and grounding, discourse markers, and the given-new contract alongside other theories discovered while trying to understand the data. These theories, together with a detailed statistical analysis of the training dataset, allowed us to design the features we considered helpful for building a system that can indicate persona-fact relevance. The features are: agency phrases, possessive phrases, discourse markers, negation, interjection, emotion polarity, temporal expressions, and position in the dialogue. With these features we ran four experiments using a linear SVM model: the first was a baseline using only the text utterances vectorized with BOW; the second used one-hot vectors that combined the presence of the linguistic features with the categorical ones; the third combined the second experiment with linguistic text features represented as n-grams; and the fourth was a hybrid that combined experiments 1 and 3 (BOW, structured cues and n-grams). The fourth experiment produced the best SVM model.

Using the model from the third experiment (structured cues and n-grams), the feature ablation showed that some features are strong indicators of whether an utterance contains relevant information. Among them, agency phrases and possessive phrases were the most informative, and removing them caused a significant drop in the model's performance. This indicates that these two features capture utterances expressing stable persona facts with humans talking about themselves, their actions, intentions, relationships, and ownership, which are important for long-term memory in multi-turn dialogue. By contrast, we observed a slight improvement in the model's performance when emotion polarity was removed, which suggests that this feature is not a reliable cue for finding relevance in a multi-turn conversation. There are several possible reasons: the feature may have been poorly extracted, VADER may not be the best system for extracting emotions from a conversation, and the emotion reflected in an utterance is context-dependent and subjective. This may indicate that emotions are not as important as other linguistic cues for signaling relevance. We also noticed a bias toward politeness, with most utterances labeled neutral (62.98% in the test dataset) or strong positive (35.07% in the test dataset) and only 1.95% strong negative in the test

dataset. This suggests that the MSC dataset consists mostly of agreeable conversations or neutral statements of persona facts, which are tied to relevance in this context.

The other features: discourse markers, interjections, temporal expressions, and position in the dialogue showed in the feature ablation that they cannot stand on their own, expressing relevance only when distributed across different pragmatic functions. Discourse markers and negation express reformulation, correction, and continuity, while temporal expressions act as anchors that mark time and connect past turns with present and future ones. But these features are weaker and behave neutrally, so we can consider them supporting cues rather than core indicators of relevance. They help organize the conversation but may not be the strongest indicators for identifying persona facts that need to be retained.

For the error analysis, we randomly selected 50 samples from the test dataset, focusing on how emotion polarity affects the model. We created a human emotion label and compared it with the VADER labels and those provided by the LLM. In doing so, we again found that there are no strong-negative labels from VADER, whereas the LLM assigned that label in a few cases, as did the human. This indicates that emotion polarity is subjective and depends on the context and tone of an utterance, not only on which words are used. At this stage we also carried out a human validation of the samples, deciding whether each sample was relevant to retain. We compared the human-validation labels with the 'gold' from the test dataset and with the predictions of the SVM model from experiment 4. The human and the 'gold' disagree in 7 of the 50 cases, the 'gold' and the SVM disagree in 30 cases, and the human and the SVM in 29. The disagreement between the human and the 'gold' indicates subjectivity in judging the samples, but also a strong overall agreement between the two annotations. During the human validation, we noticed that in 18 of the 50 cases there were annotation errors: either the crowd workers used a larger context window or they interpreted the samples differently, so that the personas do not align perfectly with the utterances. This may be one reason the SVM F1-score is penalized - an important point, because some of the SVM's errors may in fact reflect annotation errors rather than classification mistakes.

Another experiment used an LLM to indicate whether each utterance contains relevant or not relevant information worth retaining for future conversation. The results were compared in three scenarios: scenario 1, with the crowd worker annotations as the 'gold' and the LLM as the prediction; scenario 2, with the LLM as the 'gold' and the SVM model as the prediction; and scenario 3, with the LLM as the 'gold' and the crowd-worker labels as the prediction. The macro-average F1-score and the relevant-class F1-score were: scenario 1, 0.76 and 0.87; scenario 2, 0.73 and 0.84; and scenario 3, 0.76 and 0.87. The results are similar across the three scenarios, and we can argue that the relevance signal is stable across all three.

An earlier experiment in the thesis used an LLM as a judge to compare the SVM model's predictions with the 'gold' on the validation and test datasets, in terms of the macro-average F1-score. We used the predictions of the SVM from experiment 4 (combined model), and the results are similar to those we observed when we used an LLM to predict the relevance of the utterances. For the test dataset, the macro-average F1-score is 0.769. At this stage we also used BERTScore, but, as noted, it is not the best way to compute the score because it penalizes wording and paraphrasing.

Overall, our findings support the idea that relevance in long-term conversation is a phenomenon involving many factors. It depends on discourse organization, temporal expressions, persona content, and conversational context, but these do not contribute

equally in all cases. The strongest signals preserve the persona facts, while noisier cues such as emotion polarity can reduce performance. This suggests that a transparent, interpretable approach can be useful for studying relevance in long-term conversation and help personalize the LLM-human conversation.

Finally, one important note to discuss is that integrating long-term memory in the AI systems can allow the systems to adapt to the needs and preferences of the users, providing a more personalized experience. However, there are many ethical risks Lee (2024) and vulnerabilities that arise with the possibility of retaining information across sessions. One of the risks is data privacy and security, because AI models collects private information which can be accessed by third-party if there is any data leakage.

5.1 Limitations of the Study

The first limitation is that the study relies on a linear SVM model. This was a deliberate choice, made to keep the features interpretable, but non-linear models or neural networks might be better at capturing the deeper dependencies and context that could help in finding relevant persona facts in long-term conversation. A second limitation is that the error analysis was conducted on a small set (50 samples), owing to the size of the dataset. A third is the dataset itself, which contains noise, together with the fact that the MSC:PersonaSummary corpus does not cover all possible dialogue settings. In addition, some of the linguistic cues, such as emotion polarity, are context-dependent and subjective, which introduces ambiguity and leads to imperfect extraction that can affect the model's performance. The behavior of emotion polarity also suggests that VADER may not be the best solution for extracting emotions from a noisy corpus.

The fact that, in some cases, the 'gold' extracted persona is not perfectly aligned with the utterances is a further obstacle to building a fair system that can detect relevance at the utterance level, since some of the extracted persona facts draw on a wider context window in the dialogue. The utterance-level evaluation is itself a limitation. As we saw, the dataset inherited a politeness bias, with strong negative emotions outweighed by neutral and strong positive ones, which may not be representative of genuinely noisy conversations. Finally, given the time frame we were not able to reproduce the optional part of the evaluation setup discussed in the main paper for comparison: response generation with an LLM conversation model under different conditions (full dialogue history, gold summary, predicted summary from the main paper, and the summary generated by our SVM model) and to report the perplexity metric over the held-out test sessions.

Chapter 6

Conclusion

The purpose of this thesis was to find out what is relevant in human-human conversation to be retained and reused in long-term conversation. We also wanted to find out how relevance, which can be described as a 'fuzzy' term, can be detected through linguistic cues. The theoretical framework that helped us understand relevance drew on several interconnected theories: common ground, discourse markers, and given-new information. These provided a starting point for understanding why some utterances in MSC:PersonaSummary were more valuable and contained more persona facts than others, and why they are worth keeping in memory for long-term conversation.

Using the MSC:PersonaSummary data, we examined how, in a human-human multi-session conversation, persona information is established and reused across sessions after a time gap.

From a methodological perspective, we proposed a theory-driven approach that includes linguistic cues such as first-person agency, possessive phrases, discourse markers, negation, interjections, temporal expressions, emotion polarity, and position in the dialogue. The four experiments showed that combining lexical information with the linguistic cues produced the best performance on both the validation and test datasets. During the feature ablation, we found that agency phrases and possessive phrases are the most impactful features for the SVM model, while the others have little impact on their own, outside a combination with other features. The unexpected exception was the emotion polarity feature, which proved redundant and harmful to the system: the SVM model's score improves slightly when this feature is dropped. This finding from the feature ablation was confirmed by the error analysis, which specifically examined the impact of emotion polarity on the performance of the SVM system. The error analysis confirmed that emotion polarity tends to introduce noise, because the emotion labels assigned to utterances do not always align with the context and meaning of the samples we selected.

We evaluated the summaries created with our SVM model in two ways: BERTScore and an LLM as a judge. The LLM evaluation showed that the summaries preserved a good amount of the persona facts and produced higher scores than BERTScore. We argue that BERTScore may not be the best method for evaluating this type of dataset, because even when two summaries convey the same information, differences in paraphrasing and wording lower the score, as is the case for utterances versus summaries. We concluded that, for this task, the LLM is more appropriate because it ignores wording and filler words: an LLM judge can recognize a persona fact even when it is paraphrased in the summary.

The human validation, conducted on the 50 samples randomly selected from the test dataset for the error analysis, showed that relevance is context-sensitive and context-dependent: some of the extracted persona facts in the 'gold' drew on a wider conversational context and did not refer to the sample itself. We also found that some extracted personas are not entirely correct or related to the exact information given in the samples, having been interpreted differently, which suggests that relevance and persona facts are subjective.

Using an LLM to indicate which utterances are relevant or not in the test dataset, and comparing this with the SVM predictions and the 'gold' from the dataset, showed that the relevance signal is stable across the different scenarios.

With respect to the three hypotheses stated in the methodology, H1 is supported by the ablation study: utterances carrying linguistic cue features, especially agency and possessive phrases are more likely to be relevant, and removing these features causes the performance to degrade the most. H2 is supported by the four experiments: the combined full model (experiment 4), which combines the linguistic cues with the text utterance, outperforms all other experiments on both the validation and the test set, so cues and lexical information are complementary. H3 is supported by the interpretable linear SVM, which reaches a macro-F1 of about 0.76 on validation dataset and 0.75 on test dataset while still exposing transparent, interpretable feature weights.

Finally, returning to the research questions:

- *What linguistic cues in human-human interaction indicate what is 'relevant' for continuing a conversation over the long term?*
- *Can we use these linguistic cues to promote more human-like multi-turn conversations in human-LLM interaction?*

We can argue that we answered them as follows:

RQ1: Through the linguistic theories and through feature engineering and ablation studies, we showed which linguistic cues are the most important for understanding relevance in a multi-turn conversation. The research showed that agency phrases and possessive phrases are the most important for detecting relevance in long-term conversation. The reason may be that, in most cases, people talk about themselves, their preferences, likes and dislikes, and ownership. An analysis of emotion polarity showed that it is noisy, and that the way it is extracted using VADER reflects a bias toward neutral and strong positive emotion types.

RQ2: We answered this question in a methodological way, showing that linguistic cues can be used to build an interpretable relevance-selection system that may help to select and preserve memory content. For an LLM to achieve 'human-like' interaction, it needs to mimic human memory (for example, retaining and updating common ground and understanding the mechanism of given-new contract information). To evaluate the persona facts extracted by our SVM model, we used the BERTScore and LLM as a judge frameworks, which showed that the system is capable of extracting many relevant persona facts.

6.1 Future Work

Future work could extend RQ2, which would involve reproducing the experiments from the main paper. Another direction would be to apply the same linguistic cues and

theories to different datasets, to check whether these features remain informative for finding relevance in human dialogue. A further direction would be to incorporate a context window that can help capture relevance beyond the utterance level, especially given the cases in which the annotators used a wider context window in the extracted persona. Future work could also focus on improving the ambiguous features, such as emotion polarity, by combining automatic extraction with human verification to ensure the validity of the labels; the politeness bias in the dataset and the way this feature was extracted are reflected in the roughly 2% of strong negative labels. It would also be worth reconsidering the agency phrases that refer to the first person, a feature that could be made even more powerful by including the samples in which the 'I' is dropped, since human language is noisy, especially in casual conversation. Adapting a model to recognize this kind of implicit first person may be an important step toward capturing more persona facts.

Appendix A

Supplementary Material

```
======================================================================
ML EXPERIMENT: bow_baseline
======================================================================
              precision    recall  f1-score   support

not_relevant       0.66      0.69      0.67      9027
    relevant       0.83      0.80      0.81     16432

    accuracy                           0.76     25459
   macro avg       0.74      0.75      0.74     25459
weighted avg       0.77      0.76      0.76     25459
```

Figure A.1: Classification report for the first experiment (bag-of-words baseline) on the validation dataset.

```
======================================================================
ML EXPERIMENT: structured_cues
======================================================================
              precision    recall  f1-score   support

not_relevant       0.75      0.54      0.63      9027
    relevant       0.78      0.90      0.84     16432

    accuracy                           0.77     25459
   macro avg       0.76      0.72      0.73     25459
weighted avg       0.77      0.77      0.76     25459
```

Figure A.2: Classification report for the second experiment (structured cues) on the validation dataset.

```
======================================================================
ML EXPERIMENT: structured_cues_plus_cue_ngrams
======================================================================
              precision    recall  f1-score   support

not_relevant       0.66      0.71      0.68      9027
    relevant       0.83      0.80      0.82     16432

    accuracy                           0.77     25459
   macro avg       0.75      0.75      0.75     25459
weighted avg       0.77      0.77      0.77     25459
```

Figure A.3: Classification report for the third experiment (structured cues plus cue n-grams) on the validation dataset.

```
======================================================================
ML EXPERIMENT: combined_full_model
======================================================================
              precision    recall  f1-score   support

not_relevant       0.67      0.73      0.70      9027
    relevant       0.85      0.80      0.82     16432

    accuracy                           0.78     25459
   macro avg       0.76      0.77      0.76     25459
weighted avg       0.78      0.78      0.78     25459
```

Figure A.4: Classification report for the fourth experiment (combined full model) on the validation dataset.

```
======================================================================
ML EXPERIMENT: bow_baseline
======================================================================
              precision    recall  f1-score   support

not_relevant       0.63      0.69      0.66      7978
    relevant       0.84      0.80      0.82     16397

    accuracy                           0.77     24375
   macro avg       0.74      0.75      0.74     24375
weighted avg       0.77      0.77      0.77     24375
```

Figure A.5: Classification report for the first experiment (bag-of-words baseline) on the test dataset.

```
======================================================================
ML EXPERIMENT: structured_cues
======================================================================
              precision    recall  f1-score   support

not_relevant       0.71      0.54      0.61      7978
    relevant       0.80      0.89      0.84     16397

    accuracy                           0.78     24375
   macro avg       0.76      0.72      0.73     24375
weighted avg       0.77      0.78      0.77     24375
```

Figure A.6: Classification report for the second experiment (structured cues) on the test dataset.

```
======================================================================
ML EXPERIMENT: structured_cues_plus_cue_ngrams
======================================================================
              precision    recall  f1-score   support

not_relevant       0.61      0.71      0.66      7978
    relevant       0.85      0.78      0.81     16397

    accuracy                           0.76     24375
   macro avg       0.73      0.75      0.74     24375
weighted avg       0.77      0.76      0.76     24375
```

Figure A.7: Classification report for the third experiment (structured cues plus cue n-grams) on the test dataset.

```
======================================================================
ML EXPERIMENT: combined_full_model
======================================================================
              precision    recall  f1-score   support

not_relevant       0.63      0.73      0.68      7978
    relevant       0.86      0.80      0.83     16397

    accuracy                           0.77     24375
   macro avg       0.75      0.76      0.75     24375
weighted avg       0.78      0.77      0.78     24375
```

Figure A.8: Classification report for the fourth experiment (combined full model) on the test dataset.

Appendix B

Supplementary Material

Data Statement (Bender and Friedman (2018)): Multi-Session Chat Dataset (MSC)

The MSC dataset[1] appeared in the research paper published by Facebook AI titled 'Beyond Goldfish Memory: Long-Term Open-Domain Conversation'.

In the thesis, we are using a subset of the MSC dataset called MSC:PersonaSummary, which provides the raw dialogue and the summary of the sessions as per utterance extracted personas.

The dataset is available as a task in ParlAI[2], but also available to downloaded directly in JSON format for analysis: `http://parl.ai/downloads/msc/msc_v0.1.tar.gz`.

A. Curation Rationale

The MSC dataset was designed to simulate long-term human conversation that takes place over time (an episode), where the interlocutors reengage in the conversation and discuss topics that were already established in the previous sessions or they introduce new topics. An episode consists of a number of sessions (3-5) between the speakers simulating the same personas in each session, where each session is separated by a temporal gap. The MSC:PersonaSummary consists of the raw dialogue as text utterances, along with the extracted persona fact for each utterance when considered relevant for the future conversation. These extracted persona facts carries important aspects and facts about the interlocutor's persona, and are considered as gold labels signaling when the text utterance is relevant for the future conversation. Eventually, the gold summaries per session are created by combining these extracted persona facts across the session. As a result, this dataset is particularly suited for studying and researching the relevance and memory concepts in long-term conversations. The utterances in the sessions are annotated by the crowd workers. The temporal gaps that appears between the sessions in the dataset are artificially simulated.

[1] `https://parl.ai/projects/msc/`
[2] `https://github.com/facebookresearch/ParlAI`

B. Language Variety

The MSC dataset is in English and the language is casual. The conversation tries to mimic the conversation that happens in the real world on different social platforms. This kind of language can be characterized as informal and it contains filler markers, discourse markers and variable punctuation marks etc.

C. Speaker Demographic

The original paper that introduced the MSC dataset doesn't mention information about the age, gender, race/ethnicity, native language, socioeconomic status, presence of disordered speech etc. about the people that participated in the collection of this dataset.

The crowd workers were asked not to share personal information due to the privacy concerns and in order to ensure the diversity of the dialogues. They were asked to play a role that was assigned (persona), which shouldn't reflect aspect from their own lives. The emphasis was on persona-play setting that can lead to multi-turn conversations.

D. Annotator Demographic

In the MSC dataset, the crowd workers that participated in the dialogue with persona-roles for creating the dataset with multi-turn utterances were also the ones that annotated the utterances. The annotation can be described as an extracted persona fact for each utterance where the annotator feels the utterance was relevant for the future conversation (Gold Labels). The summaries created by combining these extracted persona facts per session were used as a context before the next session by the speakers in order to drive the conversation further. There were no demographic aspects revealed about them due to privacy concerns.

E. Speech Situation

The dataset consists of typed text similar to what happens in real world on different chat interfaces. These chats were conducted over multiple sessions over an episode separated artificially by hours or days. In each session, the speakers continue to interact and share aspects about the personas that were assigned. The crowd workers knew that they were interacting with another human and that the conversations are being used to create a dataset. They were asked to maintain the persona consistency in the conversation.

F. Text Characteristics

The dataset contains a multi-session dialogue that focuses on persona content and what can be retained as relevant facts for the future conversations. There are references about the topics discussed in the previous sessions, topic continuation, explicit summaries that can be considered important for the personas. The current thesis focuses on finding linguistic cues that can provide an insight on what is relevant in human dialogue. The MSC:PersonaSummary is appropriate to use because the utterances contains discourse markers, negation, first persona references, temporal expressions, emotions polarity etc. along with a clear indication on whether it was relevant to extract fact. The MSC dataset consist of episodes over 5 sessions, with each session having up to 14 utterances.

G. Other Information

The MSC:PersonaSummary is split into three parts: train, validation and test. The train dataset consists 4000 episodes of 3 sessions each, while the validation and test dataset have 500 episodes with 4 sessions each. This is different than the MSC dataset where there is an extra session for each data split - this is because the annotators were only asked to annotate the sessions that will be used later for the next session. As a result, the last sessions for each data split were not annotated.

LLM Prompts for summary session evaluation

The prompts were generated with the help of Codex and Gemini.

Summary-evaluation prompts (validation set)

PROMPT_1 = """Compare the model-generated session summary against the gold persona summary for each session_key in give CSV dataset.

Report the score for each session key as the fraction of gold persona facts captured in the model generated session summary.

The scoring scale:

0 = captures none of the gold persona facts

1 = captures all gold persona facts

Step 1: Identify and count the distinct persona facts in the Gold Summary.

Step 2: Check the Model Summary to see how many of those specific Gold facts were successfully captured.

Step 3: Calculate the coverage_score as a decimal (captured_facts / total_gold_facts). If there are 0 gold facts, the score is 1.0.

Please report the results for all given session keys in CSV format with the columns: session_key: str, total_gold_facts: int, captured_facts: int, coverage_score: float.

CSV Dataset:

"""

FOLLOW_UP_PROMPT_1 = """Excellent work on the previous step. Now, using the exact same CSV dataset and the analysis you just performed, calculate the Factual Precision score for each session key.

The Precision score represents the fraction of model-generated facts that are actually true and supported by the gold persona summary.

0 = all model facts are hallucinations | 1 = all model facts are verified by the gold summary

Step 1: For each session, identify and count the total number of distinct persona facts stated in the Model Summary (total_model_facts).

Step 2: Reuse the "captured_facts" value you successfully identified in the previous prompt. (These are the true, verified facts).

Step 3: Calculate the precision_score as a decimal (captured_facts / total_model_facts). If total_model_facts is 0, the score is 1.0.

Please report the results in a valid CSV format with the exact columns:

session_key,total_model_facts,captured_facts,precision_score

"""

NEXT_FOLLOW_UP_PROMPT_1 = "The scores here are suspiciously all 1.0, this is not correct. Please revise and calculate the scores correctly for all 500 session keys."

Summary-evaluation prompts (test set)

PROMPT_2 = """Compare the model-generated session summary against the gold persona summary for each session_key in the given CSV dataset.

Report both the Recall score (coverage of ground truth) and the Precision score (factual accuracy/lack of hallucinations) for each session key.

The scoring scales (0.0 to 1.0):

Recall: 0 = captures none of the gold facts | 1 = captures all gold facts

Precision: 0 = all model facts are hallucinations | 1 = all model facts are verified by gold summary

Step 1: Identify and count the distinct persona facts in the Gold Summary (total_gold_facts).

Step 2: Identify and count the distinct persona facts stated in the Model Summary (total_model_facts).

Step 3: Compare the two sets of facts to find the valid overlap. Count how many facts from the model summary are explicitly supported or logically entailed by the gold summary (matched_facts).

Step 4: Calculate the recall_score as a decimal (matched_facts / total_gold_facts). If total_gold_facts is 0, the score is 1.0.

Step 5: Calculate the precision_score as a decimal (matched_facts / total_model_facts). If total_model_facts is 0, the score is 0.0.

CRITICAL: Please verify your work to make sure it is correct, especially if you see a lot of suspicious scores of 1.0 or 0.0.

Please report the results for all given session keys in a valid CSV format with the exact columns:

session_key,total_gold_facts,total_model_facts,matched_facts,recall_score,precision_score

CSV Dataset:

"""

FOLLOW_UP_PROMPT_2 = "Some of the precision and recall score seems suspiciously 1.0, are you sure this is all correct? Verify your work and go step by step."

Prompt Emotion Classification

You are an expert linguistic annotator. Your task is to classify the following 50 samples of conversation with three categories of emotions: neutral, strong positive, and strong negative. For each sample assign one out of the three labels.

Definitions for the labels:

Neutral - for utterances that don't provide a clear emotional tone, or is mostly informative or expresses general opinions about a situation, and it lacks emotional intensity.

Strong positive - for utterances that indicates enthusiasm, joy, positive life experiences.

Strong negative - for utterances that indicates a negative life event or an opinion about a sad situation, or clear dislike, frustration and sadness.

Follow the guidelines:

Take into the consideration the overall emotion of the sample, and not just the individual words.

If there are not strong emotions, choose neutral.

If strong positive and strong negative are both present, pick the most dominant one.

Output one label for each sample.

The samples will be from 1 to 50, and there is no context connection between themselves.

Write the output in CSV format with text,emotion.

Samples:

Really?! I didn't know that! I sure hope you will be able to make it, especially now that you have offered to help with the meat ;)

Yes, once. It was crazy we had to shut the whole bench down for the day. There was like a group of them.

They're doing well. They're telling me about all this pop music that they like but I don't know any of the bands. Do you know any pop bands?

Yes, there is no way I could have left him, I am happy to give him a home. Have you ever rescued an animal?

haha definitely! I was thinking about getting more cats, but I am worried I might become one of those crazy cat people.

Well It keeps me bust because I clean the house too. The kids are fun but one of them is a little crazy. He bit me yesterday!

Haha no clue, but I am pretty excited about it. It will definitely help me to feel closer to my neighbors.

Yeah...if you don't like the movies marvel makes then you have been pretty much out of luck the past 12 years

Oh right, I guess she is at that age! She will grow out of it and you two will be best friends again when she comes to her senses!

Picanha is a sirloin cut that has a nice fat cap on top. It's usually slowly roasted so that the fat can render and the meat becomes tender. Wow I think I'm writing poetry

I'm so glad to hear that! Is he your first dog or have you had other pets before?

How is the wedding planning? I hope the mother in law has calmed down a little bit?

Oh my, that sounds absolutely beautiful! I am not very artistic myself. What kind of paint medium did you use? And, how long did the painting take you to complete?

I think hiking is more my style but I will look into the donkey's. Did you travel with anyone when you went?

I know you're a seasoned cook cat skater and all, but you better be wearing a helmet too. Have to lead by example with kids.

Did you find a good car on one of those auto websites I sent to you?

Swimming feels so good on my knees. They get so sore. Have you ever tried getting your dog to swim with you?

They'll teach you everything there much better than I could haha, but just know that it's not as bad as it looks! Just remember to enjoy yourself and that an accident is extremely unlikely.

Eek, I've never done anything to an animal like that. To be honest, the only fish I've ever had was a Fish Filet from McDonalds. I feel like I'll probably be a little

squeamish, at least at first.

I like more modern rock definitely am more into bluesy grassroots type stuff. Maybe we could start a band!

Now im jealous haha. Going solo?

Oh, we're not living here, we just wanted to make a trip out and see it. The parking is a bit difficult with my Subaru, however.

Of course! Then you could let me in on all your new tasty vegan recipes. I hardly know my way around the kitchen as it is.

Yes, that seems like a good deal. You get all that in one.

That sounds good! I usually order the original chicken sandwich. I think they make it really well.

I don't judge, so you can leave the house as it is if you want! I don't like cherries but I think i would like to get some pears from your orchard as well.

Is there anything I can do to help? I've been in that position before and I can only imagine that the down swing is very rough financially.

I think upper arm, wrist, and behind the shoulder are pretty common. Definitely stay away from the ribs unless you enjoy pain.

I fault the car, but the claw won.

That would be awesome. I've never seen a wolf in the wild either, but they are definitely beautiful animals. On one of the trails I jog on sometimes, there are a lot of coyotes. I saw one scamper down the trail and roam the nearby neighborhood without a care in the world. Must be nice to be carefree.

The foundations are actually all already in place, basically a hardcore foundation.

That's too bad. I love re-watching favorites too, and even seeing new director's cuts. I didn't know they are coming out with a new Cruella movie from Disney, does it look interesting to you?

that sounds sweet. I wished I was mechanically inclined and could work on cars. That would be a nice job to have.

Haha I know right! Aw man I miss school! That is awesome tho!

A loss like that is so hard for family and friends. I'm sure she will appreciate you trying to understand how she feels and taking the time to read it. Was it helpful?

I'm so glad that you liked it! I watched it only once but I've thought about watching it again. Well if you tried one of mine, maybe I'll watch one of yours.

I hope she is nicer to her kids. Luckily she doesn't have a daughter to call fat. How old are your nephews? They must be so cute!

Yes! I actually did the west highland way hike. It was so beautiful! That trip made me want to adventure even more

Oh I'm sorry to hear about that.

I'm good. What kind of furniture are you getting?

I love it here. Would love to see greece someday.

That sounds pretty good, I doubt you wear your running gear to the office though!

Oh ive been enjoying this beautiful weather by going on some hikes and fishing.

I know what you mean. I never have any alone time. What will we do when we have freedom?

Oh yes, I have a big family. My mom has 7 siblings and each of them have many kids as well. I think I have about 28 cousins from my mom's side. Do you have a big family?

I find them interesting in an almost historical sense - why people follow certain religious ideas, for example

I find it so calming. One of the reasons I love San Diego. The Gaslight District is really fun, too. How long is your drive going to be?

Did you end up getting some exercise? Are you feeling better now?

Maybe working for international aid or something like that. I'm not sure yet, I should start looking!

I think so- it's kind of like the Pacific Coast highway of Scotland right?

Prompt Utterance Relevance Prediction

The prompts were generated with the help of Claude (20-06-2026).

RELEVANCE_SYSTEM_INSTRUCTIONS = """You are an expert linguistic annotator. For each utterance in a human-human conversation, decide whether it contains a PERSONA FACT about the speaker that is worth storing in long-term memory for use in future conversations.
You label utterances from a two-person conversation (Multi-Session Chat). For each utterance, output True if it states a PERSONA FACT about its own speaker: self-disclosed, reasonably durable information about them or their life. Otherwise output False.
Rules:
- The fact must be about the speaker of that utterance. "My wife teaches" → True for that speaker; "Your job sounds fun" → False. Speakers alternate each turn.
- Use earlier turns only to interpret an utterance, never to relabel it.
- An utterance mixing a question with self-disclosure → True.
- A restated fact is still True.
Output ONLY CSV: header 'initial_data_id, session_id,turn_index,llm_prediction', then exactly one row per input utterance, with the same initial_data_id, session_id and turn_index, and llm_prediction True or False. No prose, no code fences.
Example output:
initial_data_id,session_id,turn_index,llm_prediction
test_567,1,0,False
test_567,1,1,True
"""

RELEVANCE_PROMPT = """The following are the sessions of a human-human conversation between two speakers as CSV dataset. Label every utterance below as per system instructions. Return the CSV (initial_data_id,session_id,turn_idx,llm_prediction).
CSV Dataset:
"""

Sampled Utterances

Table B.1: The 50 randomly sampled utterances with their human validation, 'gold' relevance (extracted persona), SVM predictions, and VADER, human, and LLM emotion labels.

Initial data id	Text	Human validation	Gold Label	Best SVM model (combined full model)	SVM model with full set of features	SVM model with all features minus emotion	VADER emotion	Human emotion	LLM emotion

14262 test_ 288	Really?! I didn't know that! I sure hope you will be able to make it, especially now that you have offered to help with the meat ;)	Not Relevant	Not Relevant	Not Relevant	Relevant	Not Relevant	Strong Positive	Strong Positive	Strong Positive
11209 test_ 806	Yes, once. It was crazy we had to shut the whole bench down for the day. There was like a group of them.	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Neutral	Strong Negative	Neutral
15135 test_ 445	They're doing well. They're telling me about all this pop music that they like but I don't know any of the bands. Do you know any pop bands?	Relevant	Relevant	Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral
8952 test_ 407	Yes, there is no way I could have left him, I am happy to give him a home. Have you ever rescued an animal?	Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Strong Positive	Strong Positive	Strong Positive

15835 test_ 593	haha definitely! I was thinking about getting more cats, but I am worried I might become one of those crazy cat people.	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Strong Positive	Neutral	Strong Positive
22374 test_ 698	Well It keeps me bust because I clean the house too. The kids are fun but one of them is a little crazy. He bit me yesterday!	Relevant	Relevant	Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral
20545 test_ 364	Haha no clue, but I am pretty excited about it. It will definitely help me to feel closer to my neighbors.	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Strong Positive	Strong Positive	Strong Positive
16134 test_ 644	Yeah...if you don't like the movies marvel makes then you have been pretty much out of luck the past 12 years	Not Relevant	Relevant	Relevant	Not Relevant	Relevant	Strong Positive	Neutral	Neutral

22498 test_ 715	Oh right, I guess she is at that age! She will grow out of it and you two will be best friends again when she comes to her senses!	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Strong Positive	Neutral	Strong Positive
15308 test_ 473	Picanha is a sirloin cut that has a nice fat cap on top. It's usually slowly roasted so that the fat can render and the meat becomes tender. Wow I think I'm writing poetry	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Neutral	Neutral	Strong Positive
23826 test_ 908	I'm so glad to hear that! Is he your first dog or have you had other pets before?	Not Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Neutral	Strong Positive	Strong Positive
12885 test_ 56	How is the wedding planning? I hope the mother in law has calmed down a little bit?	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Strong Positive	Neutral	Neutral

23377 test_ 842	Oh my, that sounds absolutely beautiful! I am not very artistic myself. What kind of paint medium did you use? And, how long did the painting take you to complete?	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Strong Positive	Strong Positive	Strong Posi- tive
10300 test_ 657	I think hiking is more my style but I will look into the donkey's. Did you travel with anyone when you went?	Relevant	Relevant	Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral
14727 test_ 379	I know you're a seasoned cook cat skater and all, but you better be wearing a helmet too. Have to lead by example with kids.	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Neutral	Neutral	Neutral
20966 test_ 432	Did you find a good car on one of those auto websites I sent to you?	Not Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral

7385 test_ 120	Swimming feels so good on my knees. They get so sore. Have you ever tried getting your dog to swim with you?	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral
10291 test_ 655	They'll teach you everything there much better than I could haha, but just know that it's not as bad as it looks! Just remember to enjoy yourself and that an accident is extremely unlikely.	Not Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Strong Positive	Neutral	Strong Positive
23955 test_ 927	Eek, I've never done anything to an animal like that. To be honest, the only fish I've ever had was a Fish Filet from McDonalds. I feel like I'll probably be a little squeamish, at least at first.	Relevant	Relevant	Relevant	Not Relevant	Relevant	Neutral	Strong Negative	Strong Negative

19562 test_ 162	I like more modern rock definitely am more into bluesy grassroots type stuff. Maybe we could start a band!	Relevant	Relevant	Relevant	Not Relevant	Relevant	Strong Positive	Neutral	Strong Positive
9628 test_ 530	Now im jealous haha. Going solo?	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Neutral	Neutral	Neutral
22074 test_ 645	Oh, we're not living here, we just wanted to make a trip out and see it. The parking is a bit difficult with my Subaru, however.	Relevant	Relevant	Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral
23148 test_ 816	Of course! Then you could let me in on all your new tasty vegan recipes. I hardly know my way around the kitchen as it is.	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Neutral	Neutral	Strong Positive
16864 test_ 763	Yes, that seems like a good deal. You get all that in one.	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Strong Positive	Neutral	Neutral

15631 test_ 551	That sounds good! I usually order the original chicken sandwich. I think they make it really well.	Relevant	Relevant	Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral
22027 test_ 640	I don't judge, so you can leave the house as it is if you want! I don't like cherries but I think i would like to get some pears from your orchard as well.	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Strong Positive	Neutral	Neutral
22549 test_ 723	Is there anything I can do to help? I've been in that position before and I can only imagine that the down swing is very rough financially.	Relevant	Relevant	Relevant	Not Relevant	Relevant	Neutral	Strong Negative	Strong Negative
18082 test_ 938	I think upper arm, wrist, and behind the shoulder are pretty common. Definitely stay away from the ribs unless you enjoy pain.	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Neutral	Neutral	Neutral

1673 test_ 234	I fault the car, but the claw won.	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Strong Positive	Neutral	Neutral
11104 test_ 786	That would be awesome. I've never seen a wolf in the wild either, but they are definitely beautiful animals. On one of the trails I jog on sometimes, there are a lot of coyotes. I saw one scamper down the trail and roam the nearby neighborhood without a care in the world. Must be nice to be carefree.	Relevant	Relevant	Relevant	Not Relevant	Relevant	Strong Positive	Strong Positive	Strong Positive
15933 test_ 609	The foundations are actually all already in place, basically a hardcore foundation.	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Neutral	Neutral	Neutral

11894 test_ 901	That's too bad. I love re-watching favorites too, and even seeing new director's cuts. I didn't know they are coming out with a new Cruella movie from Disney, does it look interesting to you?	Relevant	Relevant	Relevant	Not Relevant	Relevant	Strong Positive	Strong Positive	Neutral
6920 test_ 50	that sounds sweet. I wished I was mechanically inclined and could work on cars. That would be a nice job to have.	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral
4053 test_ 650	Haha I know right! Aw man I miss school! That is awesome tho!	Relevant	Relevant	Relevant	Not Relevant	Relevant	Strong Positive	Strong Positive	Strong Positive

10678 test_ 721	A loss like that is so hard for family and friends. I'm sure she will appreciate you trying to understand how she feels and taking the time to read it. Was it helpful?	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Strong Positive	Strong Negative	Strong Negative
24140 test_ 957	I'm so glad that you liked it! I watched it only once but I've thought about watching it again. Well if you tried one of mine, maybe I'll watch one of yours.	Not Relevant	Relevant	Relevant	Not Relevant	Relevant	Strong Positive	Strong Positive	Strong Positive
18827 test_ 58	I hope she is nicer to her kids. Luckily she doesn't have a daughter to call fat. How old are your nephews? They must be so cute!	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Strong Positive	Neutral	Strong Negative

13217 test_ 103	Yes! I actually did the west highland way hike. It was so beautiful! That trip made me want to adventure even more	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Strong Positive	Strong Positive	Strong Positive
6268 test_ 948	Oh I'm sorry to hear about that.	Not Relevant	Not Relevant	Not Relevant	Relevant	Not Relevant	Neutral	Neutral	Strong Negative
1046 test_ 142	I'm good. What kind of furniture are you getting?	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Neutral	Neutral	Neutral
4683 test_ 739	I love it here. Would love to see greece someday.	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Strong Positive	Strong Positive	Strong Positive
6148 test_ 935	That sounds pretty good, I doubt you wear your running gear to the office though!	Not Relevant	Not Relevant	Relevant	Relevant	Not Relevant	Neutral	Neutral	Neutral
7853 test_ 184	Oh ive been enjoying this beautiful weather by going on some hikes and fishing.	Relevant	Relevant	Relevant	Not Relevant	Relevant	Strong Positive	Strong Positive	Strong Positive
10161 test_ 637	I know what you mean. I never have any alone time. What will we do when we have freedom?	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Strong Positive	Neutral	Strong Negative

22819 test_ 769	Oh yes, I have a big family. My mom has 7 siblings and each of them have many kids as well. I think I have about 28 cousins from my mom's side. Do you have a big family?	Relevant	Relevant	Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral
14352 test_ 310	I find them interesting in an almost historical sense - why people follow certain religious ideas, for example	Relevant	Relevant	Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral
17017 test_ 784	I find it so calming. One of the reasons I love San Diego. The Gaslight District is really fun, too. How long is your drive going to be?	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Strong Positive	Strong Positive	Strong Positive
21989 test_ 634	Did you end up getting some exercise? Are you feeling better now?	Not Relevant	Not Relevant	Not Relevant	Relevant	Not Relevant	Neutral	Neutral	Neutral

17995 test_ 924	Maybe working for international aid or something like that. I'm not sure yet, I should start looking!	Relevant	Relevant	Not Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral
13219 test_ 103	I think so- it's kind of like the Pacific Coast highway of Scotland right?	Not Relevant	Relevant	Relevant	Not Relevant	Relevant	Neutral	Neutral	Neutral

References

S. Bae, D. Kwak, S. Kang, M. Y. Lee, S. Kim, Y. Jeong, H. Kim, S.-W. Lee, W. Park, and N. Sung. Keep me updated! memory management in long-term conversations. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022.

E. M. Bender and B. Friedman. Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6:587–604, 2018. doi: 10.1162/tacl_a_00041. URL `https://aclanthology.org/Q18-1041/`.

E. M. Bender and A. Lascarides. *Linguistic Fundamentals for Natural Language Processing II: 100 Essentials from Semantics and Pragmatics*. Synthesis Lectures on Human Language Technologies. Morgan & Claypool Publishers, 2019. ISBN 978-3-031-01044-6. doi: 10.2200/S00935ED1V02Y201907HLT043. URL `https://doi.org/10.2200/S00935ED1V02Y201907HLT043`.

E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, pages 610–623, 2021.

P. Brown and S. C. Levinson. *Politeness: Some Universals in Language Usage*. Cambridge University Press, Cambridge, 1987.

T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei. Language models are few-shot learners. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc., 2020. URL `https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf`.

H. H. Clark and S. E. Brennan. Grounding in communication. In L. B. Resnick, J. M. Levine, and S. D. Teasley, editors, *Perspectives on Socially Shared Cognition*, pages 127–149. American Psychological Association, Washington, DC, 1991.

H. H. Clark and S. E. Haviland. Comprehension and the given-new contract. In R. O. Freedle, editor, *Discourse Production and Comprehension*, pages 1–40. Ablex, Norwood, NJ, 1977.

C. Cortes and V. Vapnik. Support-vector networks. *Machine learning*, 20(3):273–297, 1995.

K. Drummond and R. Hopper. Back channels revisited: Acknowledgment tokens and speakership incipiency. *Research on Language and Social Interaction*, 26(2):157–177, 1993.

T. Givón. Negation in language: Pragmatics, function, ontology. In P. Cole, editor, *Syntax and Semantics, Volume 9: Pragmatics*, pages 69–112. Academic Press, New York, 1978.

H. P. Grice. Logic and conversation. In P. Cole and J. L. Morgan, editors, *Syntax and Semantics, Volume 3: Speech Acts*, pages 41–58. Academic Press, New York, 1975.

M. Honnibal, I. Montani, S. Van Landeghem, and A. Boyd. spaCy: Industrial-strength Natural Language Processing in Python. 2020. doi: 10.5281/zenodo.1212303.

C. J. Hutto and E. Gilbert. VADER: A parsimonious rule-based model for sentiment analysis of social media text. In *Proceedings of the Eighth International AAAI Conference on Weblogs and Social Media (ICWSM)*, pages 216–225, 2014.

J. Jang, M. Boo, and H. Kim. Conversation chronicles: Towards diverse temporal and relational dynamics in multi-session conversations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

G. Jefferson. Notes on a systematic deployment of the acknowledgement tokens 'yeah' and 'mm hm'. *Papers in Linguistics*, 17(2):197–216, 1984.

L. Krause. *Contextualising Conversational AI.* Phd-thesis - research and graduation internal, Vrije Universiteit Amsterdam, Oct. 2025.

E. Lee. Towards ethical personal ai applications: Practical considerations for ai assistants with long-term memory, 2024. URL `https://arxiv.org/abs/2409.11192`.

S. C. Levinson. *Pragmatics.* Cambridge University Press, Cambridge, 1983.

L. Liu, X. Yang, Y. Shen, B. Hu, Z. Zhang, J. Gu, and G. Zhang. Think-in-memory: Recalling and post-thinking enable llms with long-term memory, 2023. URL `https://arxiv.org/abs/2311.08719`.

P.-E. Mazaré, S. Humeau, M. Raison, and A. Bordes. Training millions of personalized dialogue agents. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2775–2779, 2018.

A. H. Miller, W. Feng, A. Fisch, J. Lu, D. Batra, A. Bordes, D. Parikh, and J. Weston. Parlai: A dialog research software platform, 2018. URL `https://arxiv.org/abs/1705.06476`.

F. Pedregosa et al. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

A. Qamar, J. Tong, and R. Huang. Do LLMs understand dialogues? a case study on dialogue acts. In W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 26219–26237, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. doi: 10.18653/v1/2025.acl-long.1271. URL `https://aclanthology.org/2025.acl-long.1271/`.

E. A. Schegloff. The routine as achievement. *Human Studies*, 9(2-3):111–151, 1986.

D. Schiffrin. *Discourse Markers*. Studies in Interactional Sociolinguistics. Cambridge University Press, Cambridge, 1987.

D. Sperber and D. Wilson. *Relevance: Communication and Cognition*. Blackwell, Oxford, 2nd edition, 1995.

Tools for Clear Speech. Content and Function Words, 2026. URL `https://tfcs.baruch.cuny.edu/content-and-function-words`.

Q. Wang, Y. Fu, Y. Cao, S. Wang, Z. Tian, and L. Ding. Recursively summarizing enables long-term dialogue memory in large language models, 2025. URL `https://arxiv.org/abs/2308.15022`.

Wes Mckinney. Data Structures for Statistical Computing in Python. In Stéfan van der Walt and Jarrod Millman, editors, *Proceedings of the 9th Python in Science Conference*, pages 51 – 56, 2010. doi: 10.25080/Majora-92bf17d1-00a.

Y. Wu, S. Liang, C. Zhang, Y. Wang, Y. Zhang, H. Guo, R. Tang, and Y. Liu. From human memory to ai memory: A survey on memory mechanisms in the era of llms, 2025. URL `https://arxiv.org/abs/2504.15965`.

J. Xu, A. Szlam, and J. Weston. Beyond goldfish memory: Long-term open-domain conversation. In S. Muresan, P. Nakov, and A. Villavicencio, editors, *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5180–5197, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.356. URL `https://aclanthology.org/2022.acl-long.356/`.

S. Zhang, E. Dinan, J. Urbanek, A. Szlam, D. Kiela, and J. Weston. Personalizing dialogue agents: I have a dog, do you have pets too? In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 2204–2213, 2018.

T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi. BERTScore: Evaluating text generation with BERT. In *Proceedings of the 8th International Conference on Learning Representations (ICLR)*, 2020.